

Metoprolol: Drug information - UpToDate

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For additional information see "Metoprolol: Patient drug information" and "Metoprolol: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Kaspargo Sprinkle;
- Lopressor;
- Toprol XL

Brand Names: Canada

- AA-Metoprolol SR;
- AG-Metoprolol-L;
- APO-Metoprolol;
- APO-Metoprolol Type L;
- JAMP-Metoprolol-L;
- Metoprolol-L;
- PMS-Metoprolol-L;
- PRO-Metoprolol;
- PRO-Metoprolol-L;
- RIVA-Metoprolol-L;
- TEVA-Metoprolol

Pharmacologic Category

- Antianginal Agent;
- Antihypertensive;
- Beta-Blocker, Beta-1 Selective

Dosing: Adult

Angina, chronic stable

Angina, chronic stable:

Note: For vasospastic angina, beta-blockers are not recommended; calcium channel blockers and nitrates are preferred. For nonvasospastic angina, titrate beta-blocker to relieve angina or equivalent symptoms (Ref).

Immediate release (metoprolol tartrate): **Oral:** Initial: 50 mg twice daily; may increase dose at weekly intervals to desired effect; usual dosage range: 50 to 200 mg twice daily; maximum dose: 400 mg/day.

Extended release (metoprolol succinate): **Oral:** Initial: 100 mg once daily; may increase dose at weekly intervals to desired effect; maximum dose: 400 mg/day.

Note: For microvascular angina, some experts start with 50 mg/day (divided twice daily for the immediate-release formulation or once daily for the extended-release formulation) and increase to 100 to 200 mg/day as needed to control symptoms (Ref).

Atrial fibrillation/flutter, rate control

Atrial fibrillation/flutter, rate control (off-label use):

Note: Initiate cautiously in patients with concomitant heart failure. Avoid initiating or up-titrating therapy in patients with decompensated heart failure; for unstable patients, electrical cardioversion is preferred (Ref).

Acute ventricular rate control:

IV: 2.5 to 5 mg over 2 minutes; repeat dose every 5 minutes as needed; maximum total dose: 15 mg (Ref).

Maintenance of ventricular rate control:

Immediate release (metoprolol tartrate): **Oral:** Initial: 25 mg twice daily; increase dose gradually as tolerated up to 200 mg per day in 2 to 4 divided doses to achieve ventricular rate control (Ref). For doses >200 mg/day, additional heart rate reduction may not be clinically significant (Ref).

Extended release (metoprolol succinate): **Oral:** Initial: 25 to 50 mg once daily; increase dose gradually as tolerated up to 200 mg per day in 1 or 2 divided doses to achieve ventricular rate control (Ref). For doses >200 mg/day, additional heart rate reduction may not be clinically significant (Ref).

Atrial fibrillation prevention after cardiac surgery:

Note: Initiate prior to surgery (preferentially at least 48 hours before) or postoperatively when hemodynamically stable. Continue therapy at least until the first postoperative visit in patients with no other indication for beta-blocker therapy (Ref).

Immediate release (metoprolol tartrate) : **Oral:** Initial: 25 to 50 mg twice daily; titrate based on daily evaluation of hemodynamic response to the maximally tolerated dose; maximum dose: 200 mg/day (Ref).

Extended release (metoprolol succinate): **Oral:** Initial: 50 mg once daily; titrate based on daily evaluation of hemodynamic response to the maximally tolerated dose; maximum dose: 200 mg/day (Ref).

Heart failure with reduced ejection fraction

Heart failure with reduced ejection fraction:

Note: Initiate only in stable, euvolemic patients. In hospitalized patients, volume status should be optimized and IV diuretics, IV vasodilators, and IV inotropic agents successfully discontinued prior to initiating therapy. Use caution when initiating in patients with NYHA class IV symptoms or recent heart failure exacerbation (particularly in those who required inotropes during their hospital course) (Ref).

Extended release (metoprolol succinate): **Oral:** Initial: 12.5 to 25 mg once daily; up-titrate gradually (eg, doubling the dose every ≥ 1 to 2 weeks) to a target dose of 200 mg once daily while monitoring for signs and symptoms of heart failure (Ref).

Hypertension, chronic

Hypertension, chronic (alternative agent):

Note: Recommended only in patients with specific comorbidities (eg, myocardial infarction, heart failure with reduced ejection fraction, arrhythmia) (Ref).

Immediate release (metoprolol tartrate): **Oral:** Initial: 50 mg twice daily; titrate at monthly intervals as needed based on patient response up to 200 mg/day in 2 divided doses (Ref).

Extended release (metoprolol succinate): **Oral:** Initial: 25 to 100 mg once daily; titrate at monthly intervals as needed based on patient response up to 200 mg once daily (Ref).

Migraine, prevention

Migraine, prevention (off-label use):

Note: An adequate trial for assessment of effect is considered to be at least 2 to 3 months at a therapeutic dose (Ref).

Immediate release (metoprolol tartrate) : **Oral:** Initial: 25 mg twice daily; titrate slowly (eg, every 1 to 2 weeks) based on patient response and tolerability up to 200 mg/day in divided doses (Ref).

Myocardial infarction, early treatment and secondary prevention

Myocardial infarction, early treatment and secondary prevention:

Note: An **oral** beta-blocker is recommended within the first 24 hours for most patients. Patients who did not receive a beta-blocker within 24 hours of myocardial infarction should be reevaluated for secondary prevention at a later date. The optimal duration of therapy is unknown; reassess need for long-term beta-blocker use (>1 year) if no other primary indication exists (eg, angina, arrhythmia, heart failure with reduced ejection fraction, or hypertension) (Ref).

IV:

Note: Small doses of **IV** metoprolol at the time of presentation may be considered for ST-elevation myocardial infarction (STEMI) patients with hypertension or ongoing ischemia if no contraindications exist.

Initial: 5 mg; repeat dose every 5 minutes for up to 3 doses as needed based on heart rate and blood pressure; maximum total dose: 15 mg; begin oral therapy 15 to 30 minutes after the last IV dose (Ref).

Oral:

Immediate release (metoprolol tartrate): Initial: 25 to 50 mg every 6 to 12 hours in the acute setting; may consider a lower starting dose of 12.5 mg every 6 to 12 hours when there is concern for adverse effects; for outpatients, transition to twice-daily dosing of metoprolol tartrate (immediate release) or to daily metoprolol succinate (extended release); titrate dose based on heart rate and blood pressure as tolerated up to a maximum dose of 200 mg/day (Ref).

Extended release (metoprolol succinate): Initial: 25 to 50 mg once daily; titrate dose based on heart rate and blood pressure as tolerated up to 200 mg once daily (Ref).

Supraventricular tachycardia

Supraventricular tachycardia (eg, atrioventricular nodal reentrant tachycardia, atrioventricular reentrant tachycardia, focal atrial tachycardia, multifocal atrial tachycardia) (off-label use):

Note: Consider for hemodynamically stable patients if vagal maneuvers and/or adenosine are unsuccessful. Do **not** use in patients with preexcitation associated with an accessory pathway, as this can lead to ventricular arrhythmias (Ref).

Acute treatment:

IV: 2.5 to 5 mg over 2 minutes; repeat dose every 5 minutes as needed; maximum total dose: 15 mg. **Note:** Initiate cautiously in patients with concomitant heart failure. Avoid in patients with decompensated heart failure (electrical cardioversion preferred) (Ref).

Maintenance therapy:

Immediate release (metoprolol tartrate): **Oral:** Initial: 25 mg twice daily; maximum dose: 400 mg/day (Ref).

Extended release (metoprolol succinate): **Oral:** Initial: 50 mg once daily; maximum dose: 400 mg/day (Ref).

Thyrotoxicosis

Thyrotoxicosis (off-label use):

Note: For control of cardiovascular effects until euthyroidism established (Ref).

Immediate release (metoprolol tartrate): **Oral:** 25 to 50 mg every 8 to 12 hours; may also consider administering an equivalent dose of the once-daily ER formulation (metoprolol succinate) (Ref).

Ventricular arrhythmias

Ventricular arrhythmias (off-label use):

Sustained ventricular tachycardia, incessant ventricular tachycardia, or electric storm (hemodynamically stable):

Note: Beta-blockers are generally administered in addition to an antiarrhythmic drug (eg, amiodarone) for these indications. A beta-blocker is also used to reduce shocks in patients who receive an implantable cardioverter defibrillator for these indications (Ref); propranolol may be the preferred beta-blocker in these situations (Ref).

Acute ventricular tachycardia:

IV: 5 mg every 5 minutes up to 3 doses (Ref).

Prevention of ventricular arrhythmias :

Immediate release (metoprolol tartrate): **Oral:** Initial: 12.5 to 25 mg twice daily; increase as needed based on patient response; maximum dose: 200 mg/day in 2 or 3 divided doses (Ref).

Extended release (metoprolol succinate): **Oral:** 25 to 100 mg 1 to 2 times daily (Ref).

Nonsustained ventricular tachycardia or ventricular premature beats, symptomatic:

Immediate release (metoprolol tartrate): **Oral:** 50 to 200 mg/day in 2 or 3 divided doses (Ref).

Extended release (metoprolol succinate): **Oral:** 25 to 100 mg 1 to 2 times daily (Ref).

Switching dosage forms:

Switching from immediate release (metoprolol tartrate) to extended release (metoprolol succinate): The same total daily dose of metoprolol should be used. Metoprolol tartrate is typically administered in 2 to 3 divided daily doses and metoprolol succinate is administered once daily.

Switching between oral and intravenous dosage forms: In most cases, equivalent beta-blocking effect is achieved when a 2.5:1 (Oral:IV) ratio is used. However, in one bioavailability study of healthy volunteers (N=5), the range of Oral:IV conversion ratios was approximately 2:1 to 5:1 (Ref). Therefore, patient variability may exist and a specific ratio may not apply to all patients, especially if comorbid conditions are present. The estimated equivalent IV total daily dose should be divided into 4 equal doses. For example, if a patient is receiving a chronic oral dose of metoprolol tartrate 25 mg twice daily (total daily dose of 50 mg), this could translate to a range of 2.5 mg IV every 6 hours (based on a 5:1 ratio) to 5 mg IV every 6 hours (based on a 2.5:1 ratio). Recognizing that patients receiving larger chronic oral doses should not automatically be converted to a large IV dose, consideration should be given to further reducing the initial IV dose and basing subsequent doses on the clinical response (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: Mild to severe impairment: **IV, Oral:** No dosage adjustment necessary (Ref).

Hemodialysis, intermittent (thrice weekly): Dialyzable (metabolites): **IV, Oral:** No dosage adjustment necessary (Ref).

Peritoneal dialysis: **IV, Oral:** No dosage adjustment necessary (Ref).

CRRT: **IV, Oral:** No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): **IV, Oral:** No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Due to its extensive first-pass metabolism and high hepatic extraction ratio, metoprolol elimination is highly dependent on liver blood flow. Reduced liver blood flow (as observed in the setting of severe liver dysfunction, portocaval anastomosis, or shunting) may result in significantly higher bioavailability and prolonged half-life of metoprolol. These pharmacokinetic changes are further exacerbated in patients with reduced CYP2D6 activity (Ref).

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Note: IR metoprolol (metoprolol tartrate) should be used for initial dosing in patients with liver cirrhosis. ER metoprolol (metoprolol succinate), in general, should be avoided for initial dosing in patients with liver cirrhosis (Ref).

Child-Turcotte-Pugh class A: **IV, Oral:** No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class B: **IV, Oral:** Initial: Administer 50% of indication-specific dose or a lowest dose possible based on dosage form availability; titrate no more frequently than the usual indication-specific intervals to desired clinical effect; use with caution (Ref).

Child-Turcotte-Pugh class C: **IV, Oral:** In general, avoid use if clinically possible and appropriate; if use is required, initially administer 25% of indication-specific dose or a lowest dose possible based on dosage form availability; titrate no more frequently than the usual indication-specific intervals to desired clinical effect; use with caution (Ref).

Dosing: Older Adult

Refer to adult dosing. In the management of hypertension, consider lower initial doses and titrate to response (Ref).

Dosing: Pediatric

(For additional information see "Metoprolol: Pediatric drug information")

Dosage guidance:

Dosing: Dose should be individualized based on patient response. The commercially available 10 mg/mL oral solution (Lopressor) is not approved for use in pediatric patients; an extemporaneously compounded suspension may be used.

Heart failure

Heart failure: Limited data available: Children and Adolescents: Oral: Immediate release (metoprolol tartrate): Initial: 0.1 to 0.2 mg/kg/dose twice daily, may increase slowly (usually every 2

weeks) as needed, usual daily dose: 1 mg/kg/**day**; maximum daily dose: 3 mg/kg/**day** not to exceed 200 mg/**day** (Ref).

Hypertension

Hypertension: Note: Guidelines do not recommend beta-blockers as initial therapy in pediatric patients; beta-blockers should be reserved for use in patients who have contraindications to preferred agents or after ≥ 2 preferred agents have failed in patients with hypertension and chronic kidney disease, proteinuria, or diabetes mellitus (Ref).

Immediate-release tablets (metoprolol tartrate): Children and Adolescents: Oral: Initial: 0.5 to 1 mg/kg/dose (maximum initial dose: 25 mg/dose) twice daily; adjust dose based on patient response; maximum daily dose: 6 mg/kg/**day** not to exceed 200 mg/**day** (Ref).

Extended-release sprinkle capsules or tablets (metoprolol succinate): Children ≥ 6 years and Adolescents: Oral: Initial: 1 mg/kg/dose once daily (maximum initial dose: 50 mg/dose); adjust dose based on patient response; maximum daily dose: 2 mg/kg/**day** not to exceed 200 mg/**day**; higher doses have not been studied.

Postural orthostatic tachycardia syndrome

Postural orthostatic tachycardia syndrome (POTS): Limited data available: Children ≥ 5 years and Adolescents: Oral: Immediate release (metoprolol tartrate): Initial: 0.5 to 1 mg/kg/**day** in two divided doses; titrate gradually as tolerated to achieve clinical goal; maximum daily dose: 2 mg/kg/**day** (Ref).

Syncope

Syncope: Limited data available:

Children ≥ 7 years and Adolescents:

Tilt-table test: IV: 0.1 to 0.2 mg/kg; maximum dose: 10 mg/dose (Ref).

Chronic treatment: Oral: Immediate release (metoprolol tartrate): Usual dose: 1 to 2 mg/kg/**day** in two to three divided doses, titrate gradually as tolerated to achieve clinical goal; reported dose range: 0.5 to 2.8 mg/kg/**day**; maximum daily dose: 6 mg/kg/**day** (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage recommendations in the manufacturer's labeling; in adults with mild to severe impairment, no dosage adjustment necessary.

Dosing: Liver Impairment: Pediatric

Children ≥ 6 years and Adolescents: There are no specific dosage recommendations in the manufacturer's labeling. Consider initiating with reduced doses and gradual dosage titration due to extensive hepatic metabolism.

Adverse Reactions (Significant): Considerations

Bradyarrhythmias

Beta-blockers may cause **bradycardia, sinus pause, first-degree atrioventricular (AV) block, second-degree atrioventricular block, or complete heart block** (Ref). At maintenance dosing, second- or third-degree AV block are less likely (Ref). Beta-blocking agents with intrinsic sympathomimetic activity (eg, pindolol) may cause fewer AV conduction abnormalities than those without intrinsic sympathomimetic activity (eg, metoprolol) due to their partial agonist effects (Ref). In most cases (up to 72%), AV block associated with a beta-blocker will resolve upon discontinuation; however, there are reported cases of recurrent AV block and nearly 50% of patients with more severe AV block may require a permanent pacemaker (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Blockade of cardiac beta-1 adrenergic receptors results in slowed conduction and prolongation in the refractory period of the AV node. Slowing of AV conduction can lead to AV block (Ref).

Onset: Varied; one study included patients who were on beta-blocker for more than one month (Ref); however, other studies noted prolongation of the PR interval or AV nodal refractory period occurring anywhere from one dose to several days following treatment initiation (Ref).

Risk factors:

- Impaired AV node conduction or sinus node dysfunction (Ref).
- Acute myocardial infarction (MI) (especially inferior and posterior MI) or heart failure (Ref)
- Concurrent use of other agents that impair AV nodal conduction (eg, non-dihydropyridine calcium channel blockers, digoxin, ivabradine, select antiarrhythmic agents) (Ref)
- Older patients (Ref)

Bronchospasm

Selective beta-blockers (eg, metoprolol) have a lower risk of **bronchospasm** compared to non-cardioselective beta-blockers (Ref). Concurrent use of inhaled bronchodilators and/or corticosteroids are protective against beta-blocker-induced bronchospasm in patients with COPD or asthma (Ref). Bronchospasm is reversible upon discontinuation or use of bronchodilators (Ref).

Mechanism: Dose-related; related to pharmacologic action. Beta-blocking agents can lead to airway smooth muscle constriction by antagonism of beta-2 receptors (Ref).

Onset: Rapid; reports suggest that single doses or acute use are more likely to cause changes in FEV₁ compared to chronic use (Ref).

Risk factors:

- Moderate to severe COPD (Ref)
- Acute use (Ref)

CNS effects

Beta-blockers may cause reversible CNS effects such as **fatigue, sleep disturbance, insomnia**, vivid dreams, and memory impairment (Ref). Sexual disorders may occur; however, patients who require beta-blocker therapy have risk factors for erectile dysfunction (eg, coronary artery disease, heart failure) (Ref). In addition, there may be a psychosomatic component (Ref). Lipophilic beta-blockers (such as metoprolol, which is moderately lipophilic) penetrate the blood-brain barrier to a greater extent than hydrophilic beta-blockers, possibly leading to a greater incidence of CNS effects; however, other studies have refuted this theory (Ref). CNS effects generally resolve with dose reduction or discontinuation (Ref).

Mechanism: Dose-related; exact mechanism is not fully understood. Proposed mechanisms include presence of beta receptors in the brain, affinity and in some instances, inhibition of beta-blocking agents towards serotonin (5-HT) receptors in the brain (affecting mood and sleep), and beta-blocker-induced decreases in central sympathetic output (Ref). Beta-1 blockade may also impact sleep by blocking sympathetic signaling to the pineal gland, resulting in suppression of nighttime levels of melatonin (Ref). Beta-blockers may cause erectile dysfunction through decreased sympathetic nervous system output and subsequent decreases in luteinizing hormone secretion and testosterone stimulation (Ref).

Onset: Intermediate; CNS effects often occur within the first few weeks of treatment (Ref).

Risk factors:

- Higher starting doses (Ref)
- Older patients (Ref)

Potential/masking of hypoglycemia

Beta-blockers may worsen, prolong, or cause hypoglycemia (Ref). Additionally, beta-blockers may mask symptoms of hypoglycemia (tremor, irritability, palpitations), making diaphoresis the only symptom unaffected by beta-blockers (Ref). It is unclear if non-selective or selective beta-blockers are more likely to cause hypoglycemia as data are conflicting (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Beta-blockers inhibit hepatic gluconeogenesis and glycogenolysis (Ref). Beta-blockers also reduce activation of the sympathetic nervous system, therefore masking hypoglycemic symptoms that are catecholamine-mediated (Ref).

Onset: Varied; blood glucose recovery was significantly reduced following one dose or one day of therapy (Ref). In another study, episodes of severe hypoglycemia were reported over the course of 4 years (Ref).

Risk factors:

- Insulin-dependent diabetes (Ref)
- Type 2 diabetes mellitus (Ref)
- Patients who are hospitalized and not requiring basal insulin (Ref)

Withdrawal

Beta-blocker therapy should not be withdrawn abruptly, but gradually tapered to avoid acute tachycardia, hypertension, and/or ischemia in patients with underlying cardiovascular disease (Ref). Some studies have found an increase in propensity-adjusted mortality and cardiovascular events; however, one study did not find changes in infarct size and left ventricular function when beta-blocker was abruptly withdrawn in patients with myocardial infarction (Ref).

Mechanism: Dose-dependent; related to the pharmacologic action. Beta blockade causes upregulation of beta-receptors, enhanced receptor sensitivity, and decreased sympathetic nervous system response. Abrupt withdrawal leads to a transient sympathetic hyper-response (Ref). Another proposed mechanism involves increased platelet aggregability to epinephrine and thrombin (Ref).

Onset: Rapid/varied and transient; increases in heart rate and blood pressure appear 24 hours after abrupt withdrawal, peak after 48 hours, and subside after 7 days (Ref). Anginal symptoms reported to begin 12 to 24 hours after discontinuation (Ref). Development of adverse reactions also reported to occur 1 to 21 days after withdrawal (Ref).

Risk factors:

- Abrupt withdrawal in chronic users (Ref)
- Past medical history of coronary artery disease (including chronic stable angina) (Ref)
- Past medical history of hypertension (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Cardiovascular: Bradycardia (including sinus pause: 2% to 16%) (table 1) Bradycardia, hypotension (1% to 27%)

Table 1: **Metoprolol: Adverse Reaction: Bradycardia**

Drug (Metoprolol)	Placebo	Indication	Number of Patients (Metoprolol)	Number of Patients (Placebo)
2%	0.4%	Cardiac failure	1,990	2
16%	7%	Myocardial infarction	N/A	1

1% to 10%:

Cardiovascular: Arterial insufficiency (usually Raynaud type: 1%), cold extremity (1%), first-degree atrioventricular block (5%) (table 2) First Degree Atrioventricular Block, heart failure (1%), palpitations (1%), peripheral edema (1%)

Table 2: **Metoprolol: Adverse Reaction: First-Degree Atrioventricular Block**

Drug (Metoprolol)	Placebo	Indication
5%	2%	Myocardial infarction

Dermatologic: Gangrene of skin and/or subcutaneous tissues (1%), pruritus (5%), skin rash (>2% to 5%)

Gastrointestinal: Constipation (1%), diarrhea (>2% to 5%), flatulence (1%), heartburn (1%), nausea ($\leq 1\%$), stomach pain (1%), xerostomia (1%)

Nervous system: Cerebrovascular accident (1%), depression (>2% to 5%), dizziness (2% to 10%), fatigue (1% to 10%), vertigo ($\leq 2\%$)

Respiratory: Bronchospasm (1%), dyspnea ($\leq 3\%$), wheezing (1%)

Miscellaneous: Accidental injury (1%)

<1%:

Dermatologic: Alopecia

Gastrointestinal: Abdominal pain

Genitourinary: Peyronie disease

Nervous system: Anxiety

Neuromuscular & skeletal: Arthralgia, arthritis

Frequency not defined:

Cardiovascular: Claudication

Dermatologic: Exacerbation of psoriasis (Ref)

Endocrine & metabolic: Unstable diabetes, weight gain

Gastrointestinal: Vomiting

Genitourinary: Retroperitoneal fibrosis

Nervous system: Sleep disturbance (Ref), temporary amnesia

Ophthalmic: Visual disturbance

Otic: Tinnitus

Respiratory: Rhinitis

Postmarketing:

Cardiovascular: Chest pain, syncope

Dermatologic: Bullous pemphigoid (Ref), diaphoresis, erythroderma (Ref), lichenoid dermatitis (Ref), pemphigoid-like lesion (Ref)

Endocrine & metabolic: Decreased HDL cholesterol (Ref), decreased libido (Ref), increased serum triglycerides (Ref)

Genitourinary: Erectile dysfunction (Ref), impotence (Ref)

Hepatic: Hepatic insufficiency (Ref), hepatitis (Ref), increased serum alkaline phosphatase (Ref), increased serum transaminases (Ref), jaundice (Ref)

Nervous system: Confusion (Ref), drowsiness, hallucination (Ref), headache (Ref), insomnia (Ref), nervousness (Ref), nightmares (Ref), paresthesia

Neuromuscular & skeletal: Arthropathy (psoriatic) (Ref), musculoskeletal pain (Ref)

Contraindications

Hypersensitivity to metoprolol, any component of the formulation, or other beta-blockers; second- or third-degree heart block (except in patients with a functioning artificial pacemaker).

Note: Additional contraindications are formulation and/or indication specific.

IR oral solution: Severe bradycardia, cardiogenic shock; systolic BP <100, decompensated heart failure; sick sinus syndrome (except in patients with a permanent pacemaker).

IR tablets/injectable formulation:

Hypertension and angina (tablet only): Sinus bradycardia; cardiogenic shock; overt heart failure; sick sinus syndrome (except in patients with a functioning artificial pacemaker); severe peripheral arterial circulatory disorders.

Myocardial infarction (tablet and injection): Severe sinus bradycardia (heart rate <45 beats/minute); significant first-degree heart block (P-R interval ≥ 0.24 seconds) (except in patients with a functioning artificial pacemaker); systolic blood pressure <100 mm Hg; moderate to severe cardiac failure.

ER formulation: Severe bradycardia, cardiogenic shock; decompensated heart failure; sick sinus syndrome (except in patients with a functioning artificial pacemaker).

Canadian labeling: Additional contraindications (not in US labeling): Cor pulmonale; untreated pheochromocytoma; asthma and other obstructive respiratory disease (injection only); concomitant use with anesthesia agents that cause myocardial depression.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse events:

- Anaphylactic reactions: Use caution with history of severe anaphylaxis to allergens; patients taking beta-blockers may become more sensitive to repeated allergen challenges. Treatment of anaphylaxis (eg, epinephrine) in patients taking beta-blockers may be ineffective or promote undesirable effects.

Disease-related concerns:

- Diabetes: Use with caution in patients with diabetes mellitus; may potentiate hypoglycemia and/or mask signs and symptoms.

- Heart failure with reduced ejection fraction: Stabilize patients on heart failure regimen prior to initiation or titration of beta-blocker. Beta-blocker therapy should be initiated at very low doses with gradual and careful titration. Worsening heart failure or fluid retention may occur during upward titration; dose reduction and/or slower titration may be necessary. Adjustment of other medications (angiotensin-converting enzyme inhibitors and/or diuretics) may be required. Only the ER formulation is indicated for use in heart failure.

- Hepatic impairment: Use with caution in patients with hepatic impairment.

- Myasthenia gravis: Use beta-blockers with caution in patients with myasthenia gravis.

- Peripheral vascular disease (PVD) and Raynaud disease: May precipitate or aggravate symptoms of arterial insufficiency in patients with PVD and Raynaud disease. Use with caution and monitor for progression of arterial obstruction.

- Pheochromocytoma (untreated): Adequate alpha-blockade is required prior to use of any beta-blocker.
- Psoriasis: Beta-blocker use has been associated with induction or exacerbation of psoriasis, but cause and effect have not been firmly established.
- Supraventricular tachycardia (SVT): If antidromic atrioventricular reentrant tachycardia (AVRT) or pre-excited atrial fibrillation is suspected, avoid AV node-specific blocking drugs (eg, adenosine, diltiazem, verapamil, digoxin, beta-blockers). For these types of SVT enhanced antegrade conduction from atria to ventricles may occur through an accessory pathway leading to ventricular arrhythmias if the AV node is blocked. It is safe to use AV node-specific blocking drugs for orthodromic AVRT because antegrade conduction occurs through the AV node and only retrograde conduction (from ventricles to atria) occurs through the accessory pathway.
- Thyroid disease: May mask signs of hyperthyroidism (eg, tachycardia). If hyperthyroidism is suspected, carefully manage and monitor; abrupt withdrawal may exacerbate symptoms of hyperthyroidism or precipitate thyroid storm. Alterations in thyroid function tests may be observed.
- Vasospastic angina: Beta-blockers without alpha1-adrenergic receptor blocking activity should be avoided in patients with vasospastic angina since unopposed alpha1-adrenergic receptors mediate coronary vasoconstriction and can worsen anginal symptoms (Mayer 1998).

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol and/or sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [“Inactive” 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol and/or benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.

Other warnings/precautions:

- Abrupt withdrawal: Abrupt withdrawal may exacerbate underlying conditions, such as angina pectoris, sinus tachycardia, hypertension, and arrhythmias; cases of myocardial infarction have also occurred.
- Major surgery: Chronic beta-blocker therapy should not be routinely withdrawn prior to major surgery; abrupt discontinuation may result in rebound hypertension and is potentially harmful. In beta-blocker naïve patients, do not initiate therapy on the day of surgery (ACC/AHA [Jones 2025]).

Special populations:

- Bariatric surgery: Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Two small pharmacokinetic studies showed wide interpatient variability in metoprolol tartrate and metoprolol succinate exposure following bariatric surgery (Gesquiere 2015; Yska 2020). Monitor for continued efficacy and tolerability after bariatric surgery and consider switching to an alternate medication if symptoms worsen.

Warnings: Additional Pediatric Considerations

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, $>3,000$ mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms Considerations

The extended release metoprolol succinate products’ strengths are expressed as metoprolol tartrate equivalents.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule ER 24 Hour Sprinkle, Oral:

Kapsargo Sprinkle: 25 mg, 50 mg, 100 mg, 200 mg [contains corn starch]

Solution, Intravenous, as tartrate:

Generic: 5 mg/5 mL (5 mL)

Solution, Intravenous, as tartrate [preservative free]:

Generic: 5 mg/5 mL (5 mL)

Solution, Oral, as tartrate:

Lopressor: 10 mg/mL (300 mL) [contains disodium edta, sodium benzoate]

Tablet, Oral, as tartrate:

Lopressor: 50 mg [scored]

Lopressor: 100 mg [scored; contains fd&c blue #2 (indigo carm) aluminum lake]

Generic: 12.5 mg, 25 mg, 37.5 mg, 50 mg, 75 mg, 100 mg

Tablet Extended Release 24 Hour, Oral, as succinate:

Toprol XL: 25 mg

Toprol XL: 25 mg [scored]

Toprol XL: 50 mg

Toprol XL: 50 mg [scored]

Toprol XL: 100 mg

Toprol XL: 100 mg [scored]

Toprol XL: 200 mg

Toprol XL: 200 mg [scored]

Generic: 25 mg, 50 mg, 100 mg, 200 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsule ER 24 Hour Sprinkle (Kapsargo Sprinkle Oral)

25 mg (per each): \$2.38

50 mg (per each): \$2.38

100 mg (per each): \$2.88

200 mg (per each): \$4.51

Solution (Lopressor Oral)

10 mg/mL (per mL): \$1.50

Solution (Metoprolol Tartrate Intravenous)

5 mg/5 mL (per mL): \$0.08 - \$0.68

Tablet, 24-hour (Metoprolol Succinate ER Oral)

25 mg (per each): \$0.14 - \$1.28
50 mg (per each): \$0.12 - \$1.28
100 mg (per each): \$0.38 - \$2.04
200 mg (per each): \$0.30 - \$3.06

Tablet, 24-hour (Toprol XL Oral)

25 mg (per each): \$1.58
50 mg (per each): \$1.58
100 mg (per each): \$2.37
200 mg (per each): \$3.77

Tablets (Lopressor Oral)

50 mg (per each): \$3.02
100 mg (per each): \$4.52

Tablets (Metoprolol Tartrate Oral)

12.5 mg (per each): \$3.75
25 mg (per each): \$0.02 - \$0.24
37.5 mg (per each): \$1.46
50 mg (per each): \$0.02 - \$0.57
75 mg (per each): \$2.44
100 mg (per each): \$0.03 - \$0.80

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as tartrate:

Generic: 1 mg/mL (5 mL); 5 mg/5 mL (5 mL)

Tablet, Oral, as tartrate:

Generic: 25 mg, 50 mg, 100 mg

Tablet Extended Release 24 Hour, Oral:

Generic: 100 mg, 200 mg

Extemporaneous Preparations

Note: A 10 mg/mL oral solution is commercially available.

10 mg/mL Oral Suspension (ASHP Standard Concentration) (Ref)

A 10 mg/mL oral suspension may be made with metoprolol tartrate tablets and one of three different vehicles (cherry syrup; a 1:1 mixture of Ora-Sweet and Ora-Plus; or a 1:1 mixture of Ora-Sweet SF and Ora-Plus). Crush twelve 100 mg tablets in a mortar and reduce to a fine powder. Add 20 mL of the chosen vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 120 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of vehicle sufficient to make 120 mL. Label "shake well" and "protect from light." Stable for 60 days (Ref).

Administration: Adult

Oral:

Immediate release (metoprolol tartrate): Administer with or immediately following meals.

Extended release (metoprolol succinate): According to the manufacturer, it is preferable to administer with or immediately following meals; however, may administer without regard to meals (Ref). May divide tablets in half; do not crush or chew.

Bariatric surgery: Metoprolol has ER formulations, and the release characteristics may be significantly altered in an unknown manner in patients who have undergone bariatric surgery. Providers should determine if the condition being treated can be safely monitored or if a switch to an alternative formulation is necessary (Dvořáčková 2024). IR formulations are also available.

Sprinkle capsule: May be swallowed whole or the capsule may be opened and contents sprinkled on a small amount (1 teaspoonful) of soft food (eg, applesauce, pudding, or yogurt) to be used within 60 minutes (do not store for future use).

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Oral sprinkle capsules, extended release (metoprolol succinate):

Gastric tubes (eg, NG, G-tube) (≥12 French): Open capsule, add contents to an all-plastic enteral dosing syringe, and add 15 mL of purified water; gently shake syringe for ~10 seconds. Administer immediately via feeding tube, ensuring no pellets remain in the syringe (Ref).

General guidance: Hold enteral nutrition (EN) during metoprolol administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse enteral dosing syringe used for preparation with purified water and administer contents to ensure delivery of entire dose (Ref). Ensure no pellets remain in the enteral dosing syringe (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Oral tablets, immediate release (metoprolol tartrate):

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush and disperse tablet in 10 mL purified water; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: Some formulations may be film-coated; administration of film-coated metoprolol tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are dispersed sufficiently with an adequate amount of purified water prior to administration (Ref).

General guidance: Hold EN during metoprolol administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref). **Note:** To ensure adequate bioavailability, administer dose just prior to a scheduled intermittent enteral feed, or restart feeds immediately after final flush for continuous enteral feeds (Ref).

Oral tablets, extended release (metoprolol succinate):

Enteral feeding tube administration of metoprolol ER tablets is not recommended. Although metoprolol ER tablets may be cut (if they are scored), they should not be crushed. Crushing modified release dosage forms (eg, ER tablets) may result in release of excessive doses, variable serum concentrations, and risk of severe adverse effects including fatalities (Ref).

No te: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties among manufacturers.

IV: May administer undiluted by rapid infusion (IV push) over 1 minute for acute treatment. May also be administered by slow infusion (ie, 5 to 10 mg of metoprolol in 50 mL of fluid) over ~30 to 60 minutes during less urgent situations (eg, substitution for oral metoprolol).

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editor panel: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral:**Immediate release:**

Tablets (metoprolol tartrate): Administer with food or immediately after meals; swallow unchewed with a glass of water.

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush tablet(s) to a fine powder and disperse tablet in 10 mL purified water; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: Some tablets may be film-coated; administration of film-coated tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are sufficiently dispersed prior to administration (Ref).

General guidance: Hold enteral nutrition during metoprolol administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref). **Note:** To ensure adequate bioavailability, administer metoprolol tartrate immediate-release tablets just prior to a scheduled intermittent enteral feed, or restart continuous feeds immediately after final flush (Ref).

Oral solution (commercially available): Lopressor (metoprolol tartrate): **Note:** Approved for use in adults only.

Administration via feeding tube: Enteral feeding tube administration of metoprolol oral solution has not been evaluated.

Extended release (metoprolol succinate):

Sprinkle capsules: Administer without regard to meals; may be swallowed whole or the capsule may be opened and the contents sprinkled on small amount (1 teaspoonful) of soft food (eg, pudding, applesauce, yogurt) and used within 60 minutes of preparation; do not store drug-food mixture for later use.

Administration via feeding tube :

Gastric tubes (eg, NG, G-tube) (≥ 12 French): Open capsule, add contents to an all-plastic enteral dosing syringe and add 15 mL of purified water; gently shake syringe for ~10 seconds. Administer immediately via feeding tube, ensuring no pellets remain in the syringe; rinse with additional water if necessary (Ref).

General guidance: Hold enteral nutrition during metoprolol administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse syringe with additional purified water to ensure no pellets remain and entire dose is delivered (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Tablets: May be administered without regard to meals; Toprol-XL tablets are scored and may be divided; do not chew or crush the half or whole tablets; swallow whole. Do not chew, crush, or break generic nonscored extended-release tablets; swallow whole.

Administration via feeding tube: Enteral feeding tube administration utilizing metoprolol extended-release tablets is not recommended. Although metoprolol extended-release tablets may be cut (if they are scored), they should not be crushed. Crushing modified-release dosage forms (eg, extended-release tablets) may result in release of excessive doses, variable serum concentrations, and risk of severe adverse effects including fatalities (Ref).

Parenteral: Note: IV dose is much smaller than oral dose. When administered acutely for cardiac treatment, monitor ECG and blood pressure.

Children ≥ 7 years and Adolescents: Limited data available; dose was administered over 10 minutes in one study (Ref).

Storage/Stability

Injection: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light and heat.

Oral solution: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from heat. Once bottle is opened, use within 30 days.

Sprinkle capsules: Store at 20°C to 25°C (68°F to 77°F).

Tablet: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from moisture.

Use: Labeled Indications

Angina, chronic stable: Long-term treatment of angina pectoris.

Heart failure with reduced ejection fraction (ER oral formulation): Treatment of stable, symptomatic (NYHA class II or III) heart failure of ischemic, hypertensive, or cardiomyopathic origin to reduce the rate of mortality plus hospitalization in patients already receiving angiotensin-converting enzyme inhibitors, diuretics, and/or digoxin.

Hypertension, chronic: Management of hypertension.

Myocardial infarction: Treatment of hemodynamically stable acute myocardial infarction to reduce cardiovascular mortality (injection to be used in combination with metoprolol oral maintenance therapy).

Use: Off-Label: Adult

Atrial fibrillation/flutter, rate control; Atrial fibrillation prevention after cardiac surgery; Hypertrophic cardiomyopathy; Migraine, prevention; Supraventricular tachycardia (eg, atrioventricular nodal reentrant tachycardia, atrioventricular reentrant tachycardia, focal atrial tachycardia, multifocal atrial tachycardia); Thyrotoxicosis; Ventricular arrhythmias

Medication Safety Issues

Sound-alike/look-alike issues:

Lopressor may be confused with Lyrica

Metoprolol may be confused with metaproterenol, metoclopramide, metolazone, misoprostol

Metoprolol succinate may be confused with metoprolol tartrate

Toprol-XL may be confused with Tegretol, Tegretol XR, Topamax

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (adrenergic antagonist, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Older Adult: High-Risk Medication:

Metoprolol is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age). Some disease states of concern include bradycardia, heart block, and severe symptomatic aortic stenosis. Use is not recommended as monotherapy for the treatment of uncomplicated hypertension (O'Mahony 2023).

Administration issues:

Significant differences exist between oral and IV dosing. Use caution when converting from one route of administration to another.

Metabolism/Transport Effects

Substrate of CYP2C19 (Minor), CYP2D6 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acetylcholinesterase Inhibitors: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Ajmaline: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alpha2-Agonists: Beta-Blockers may increase rebound hypertensive effects of Alpha2-Agonists. This effect can occur when the Alpha2-Agonist is abruptly withdrawn. Alpha2-Agonists may increase AV-blocking effects of Beta-Blockers. Sinus node dysfunction may also be enhanced. Management: Closely monitor heart rate during treatment with a beta blocker and clonidine. Withdraw beta blockers several days before clonidine withdrawal when possible, and monitor blood pressure closely. Recommendations for other alpha2-agonists are unavailable. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amiodarone: May increase bradycardic effects of Beta-Blockers. Possibly to the point of cardiac arrest. Amiodarone may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Antidiabetic Agents: Beta-Blockers (Beta1 Selective) may increase adverse/toxic effects of Antidiabetic Agents. Specifically, beta-blockers may mask the hypoglycemic symptoms of antidiabetic agents. *Risk C: Monitor*

Antipsychotic Agents (Phenothiazines): May increase hypotensive effects of Beta-Blockers. Beta-Blockers may decrease metabolism of Antipsychotic Agents (Phenothiazines). Antipsychotic Agents (Phenothiazines) may decrease metabolism of Beta-Blockers. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Artemether and Lumefantrine: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta2-Agonists: Beta-Blockers (Beta1 Selective) may decrease bronchodilatory effects of Beta2-Agonists. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Cafedrine: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Cafedrine. *Risk C: Monitor*

Cannabis: Beta-Blockers may increase adverse/toxic effects of Cannabis. Specifically, the risk of hypoglycemia may be increased. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Cholinergic Agonists: Beta-Blockers may increase adverse/toxic effects of Cholinergic Agonists. Of particular concern are the potential for cardiac conduction abnormalities and bronchoconstriction. *Risk C: Monitor*

CYP2D6 Inhibitors (Moderate): May increase serum concentrations of Metoprolol. *Risk C: Monitor*

CYP2D6 Inhibitors (Strong): May increase serum concentrations of Metoprolol. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Dipyridamole: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Disopyramide: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may increase negative inotropic effects of Disopyramide. *Risk C: Monitor*

DOBUTamine: Beta-Blockers may decrease therapeutic effects of DOBUTamine. *Risk C: Monitor*

Dronedarone: May increase bradycardic effects of Beta-Blockers. Dronedarone may increase serum concentrations of Beta-Blockers. This likely applies only to those agents that are metabolized by CYP2D6. Management: Use lower initial beta-blocker doses; adequate tolerance of the combination,

based on ECG findings, should be confirmed prior to any increase in beta-blocker dose. Increase monitoring for clinical response and adverse effects. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

EPHEDrine (Systemic): Beta-Blockers may decrease therapeutic effects of EPHEDrine (Systemic). *Risk C: Monitor*

EPINEPHrine (Systemic): Beta-Blockers (Beta1 Selective) may decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): Beta-Blockers may increase vasoconstricting effects of Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates). *Risk C: Monitor*

Etilefrine: Beta-Blockers may decrease therapeutic effects of Etilefrine. Etilefrine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Etofylline: Beta-Blockers may decrease therapeutic effects of Etofylline. *Risk X: Avoid*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Grass Pollen Allergen Extract (5 Grass Extract): Beta-Blockers may increase adverse/toxic effects of Grass Pollen Allergen Extract (5 Grass Extract). More specifically, Beta-Blockers may inhibit the ability to effectively treat severe allergic reactions to Grass Pollen Allergen Extract (5 Grass Extract) with epinephrine. Some other effects of epinephrine may be unaffected or even enhanced (e.g., vasoconstriction) during treatment with Beta-Blockers. Management: Consider alternatives to either grass pollen allergen extract (5 grass extract) or beta-blockers in patients with indications for both agents. Canadian product labeling specifically lists this combination as contraindicated. *Risk D: Consider Therapy Modification*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hydroxychloroquine: May increase serum concentrations of Metoprolol. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Isoproterenol: Beta-Blockers may decrease therapeutic effects of Isoproterenol. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Lercanidipine: May increase hypotensive effects of Metoprolol. Metoprolol may decrease serum concentrations of Lercanidipine. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mavacamten: Beta-Blockers may increase adverse/toxic effects of Mavacamten. Specifically, negative inotropic effects may be increased. *Risk C: Monitor*

Mavorixafor: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk X: Avoid*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methacholine: Beta-Blockers may increase adverse/toxic effects of Methacholine. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mirabegron: May decrease antihypertensive effects of Metoprolol. Mirabegron may increase serum concentrations of Metoprolol. *Risk C: Monitor*

Mivacurium: Beta-Blockers may increase therapeutic effects of Mivacurium. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

NIFEdipine (Systemic): May increase hypotensive effects of Beta-Blockers. NIFEdipine (Systemic) may increase negative inotropic effects of Beta-Blockers. *Risk C: Monitor*

NIFEdipine (Topical): May increase adverse/toxic effects of Beta-Blockers. *Risk C: Monitor*

Nitrendipine: May increase therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May decrease therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease antihypertensive effects of Beta-Blockers. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Opipramol: Beta-Blockers may increase serum concentrations of Opipramol. Opipramol may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Parecoxib: May increase serum concentrations of Metoprolol. Parecoxib may decrease antihypertensive effects of Metoprolol. *Risk C: Monitor*

PENTobarbital: May increase hypotensive effects of Metoprolol. PENTobarbital may decrease serum concentrations of Metoprolol. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reserpine: May increase hypotensive effects of Beta-Blockers. Reserpine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

RifAMPin: May decrease serum concentrations of Metoprolol. *Risk C: Monitor*

Rivastigmine: May increase bradycardic effects of Beta-Blockers. Management: Consider alternatives to the use of rivastigmine and beta-blockers when possible, as product labeling states that the concomitant use of these agents is not recommended. If combined, monitor for bradycardia and syncope. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Sotagliflozin: Metoprolol may increase hypoglycemic effects of Sotagliflozin. Sotagliflozin may increase serum concentrations of Metoprolol. *Risk C: Monitor*

Succinylcholine: Beta-Blockers may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tasimelteon: Beta-Blockers may decrease therapeutic effects of Tasimelteon. Management: Consider avoiding nighttime administration of beta-blockers during tasimelteon therapy due to the potential for reduced tasimelteon efficacy. *Risk D: Consider Therapy Modification*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Theodrenaline: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Theodrenaline. *Risk C: Monitor*

Theophylline Derivatives: Beta-Blockers (Beta1 Selective) may decrease bronchodilatory effects of Theophylline Derivatives. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

White Birch Allergen Extract: Beta-Blockers may increase adverse/toxic effects of White Birch Allergen Extract. Specifically, beta-blockers may reduce the effectiveness of beta-agonists that may be required to treat systemic reactions to white birch allergen extract. *Risk X: Avoid*

Zuclopenthixol: May increase or decrease antihypertensive effects of Beta-Blockers. *Risk X: Avoid*

Food Interactions

Food increases absorption. Metoprolol serum levels may be increased if taken with food. Management: Take immediate release tartrate tablets with food; succinate can be taken with or without food.

Reproductive Considerations

Medications considered acceptable for the treatment of chronic hypertension during pregnancy may generally be used in patients trying to conceive. Metoprolol is generally not a preferred agent for use in pregnant patients (ACC/AHA [Whelton 2018]; ACOG 2019); however, use may be considered (ESC [Cífková 2020]; SOGC [Magee 2022]).

Metoprolol may be continued for the treatment of heart failure in patients planning to become pregnant; treatment should be made as part of a shared decision-making process (AHA/ACC/HFSA [Heidenreich 2022]).

Metoprolol is effective for prevention of migraines. In general, preventive treatment for migraine in patients trying to become pregnant should be avoided. Options for patients planning a pregnancy should be considered as part of a shared decision-making process. Nonpharmacologic interventions should be considered initially. When needed, preventive treatment should be individualized considering the available safety data and needs of the patient should pregnancy occur. A gradual discontinuation of preventive medications is generally preferred when the decision is made to stop treatment prior to conception (ACOG 2022; AHS [Ailani 2021]).

Erectile dysfunction and inhibition of sperm motility are noted in product labeling following use of metoprolol. As a class, outcomes from available studies evaluating beta-blockers and sexual dysfunction are inconsistent, and the negative effects on erectile function and libido are considered controversial. A clear relationship between use of beta-blockers and erectile dysfunction has not been established. Hypertension itself is associated with erectile dysfunction. Patients on a beta-blocker presenting with sexual dysfunction should be evaluated for underlying disease (Farmakis 2021; Levine 2012; Semet 2017; Terentes-Printzios 2022; Viigimaa 2020).

Pregnancy Considerations

Metoprolol and the metabolite alpha-hydroxymetoprolol cross the placenta (Lindeberg 1987; Ryu 2016).

Exposure to beta-blockers during pregnancy may increase the risk for adverse events in the neonate. If maternal use of a beta-blocker is needed, monitor fetal growth during pregnancy; monitor the newborn for 48 hours after delivery for bradycardia, hypoglycemia, and respiratory depression (ESC [Regitz-Zagrosek 2018]).

Chronic maternal hypertension is also associated with adverse events in the fetus/infant. Chronic maternal hypertension may increase the risk of birth defects, low birth weight, premature delivery, stillbirth, and neonatal death. Actual fetal/neonatal risks may be related to the duration and severity of maternal hypertension. Untreated chronic hypertension may also increase the risks of adverse maternal outcomes, including gestational diabetes, preeclampsia, delivery complications, stroke, and myocardial infarction (ACOG 2019).

The pharmacokinetics of metoprolol may be changed during pregnancy; the degree of changes may be dependent upon maternal CYP2D6 genotype (Ryu 2016).

Patients with preexisting hypertension may continue their medication during pregnancy unless contraindications exist (ESC [Regitz-Zagrosek 2018]). When treatment of hypertension is initiated during pregnancy, agents other than metoprolol may be preferred (ACOG 2019); however, use may be considered (ESC [Cífková 2020]; SOGC [Magee 2022]). Metoprolol may be continued for the treatment of heart failure in pregnant patients; treatment should be made as part of a shared decision-making process (AHA/ACC/HFSA [Heidenreich 2022]). Metoprolol may be used for the treatment of maternal ventricular arrhythmias, atrial fibrillation/atrial flutter, or supraventricular tachycardia during pregnancy; consult current guidelines for specific recommendations (ACC/AHA/HRS [Page 2016]; ESC [Regitz-Zagrosek 2018]).

In general, preventive treatment for migraine should be avoided during pregnancy. Options for pregnant patients should be considered as part of a shared decision-making process. Nonpharmacologic interventions should be considered initially. When needed, preventive treatment should be individualized considering the available safety data, the potential for adverse maternal and fetal events, and needs of the patient (ACOG 2022; AHS [Ailani 2021]). If preventive therapy is needed, beta-blockers may be considered (ACOG 2022). Based on available data, metoprolol may be used if prophylaxis of migraine is needed in pregnant patients; it should be discontinued 2 to 3 days prior to delivery to decrease the risk of adverse events to the fetus/neonate and potential reductions in uterine contraction (CHS [Pringsheim 2012]).

Breastfeeding Considerations

Metoprolol is present in breast milk.

Data related to the presence of metoprolol in breast milk are available from 3 lactating patients, 4 to 6 months postpartum. They were administered oral metoprolol after they stopped breastfeeding, but lactation was maintained via breast pumping. Metoprolol 50 mg twice daily was administered on day 1, 100 mg twice daily on days 2 and 3, and 100 mg on day 4. Blood and milk samples were obtained immediately prior to the dose on day 4, then at intervals from 3 to 12 hours later. The maximum maternal metoprolol blood concentrations ranged from 27 to 259 ng/mL, and the maximum milk concentrations ranged from 102 to 690 ng/mL (Liedholm 1981). Using the highest milk concentration reported (690 ng/mL), the estimated exposure to the breastfeeding infant would be 0.1 mg/kg/day (relative infant dose [RID] 7%) compared to a weight-adjusted maternal dose of 100 mg/day. In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

A case report also notes the presence of the α -OH-metoprolol metabolite in breast milk (Grundman 2011). Metoprolol is measurable in the serum of some breastfed infants (Sandström 1983). Adverse events were not reported in 6 infants exposed to metoprolol via breast milk (Ho 1999).

The manufacturer recommends monitoring the breastfed infant for adverse events such as bradycardia; constipation; diarrhea; and dry mouth, skin, or eyes when metoprolol is administered to a mother who is a slow metabolizer.

When treatment for hypertension is needed in a breastfeeding patient, consider use of an agent other than metoprolol (ACOG 2019). Metoprolol may be continued for the treatment of heart failure in lactating patients; treatment should be made as part of a shared decision-making process (AHA/ACC/HFSA [Heidenreich 2022]). In general, preventive treatment for migraine in lactating patients should be avoided. When needed, therapy should be individualized considering the available safety data and needs of the patient (AHS [Ailani 2021]).

Monitoring Parameters

ECG; heart rate; BP; serum glucose (in patients with diabetes); mental alertness.

Bariatric surgery: Monitor BP and for symptoms of hypotension since bariatric surgery causes an immediate decrease in BP; dose reduction or discontinuation of therapy may be necessary (Hawkins 2018; Wang 2021).

Reference Range

Blood pressure goal: May vary depending on clinical conditions, different clinical practice guidelines, and expert opinion. Refer to "Clinical Practice Guidelines" for specific treatment goals.

Mechanism of Action

Selective inhibitor of β_1 -adrenergic receptors; competitively blocks β_1 -receptors, with little or no effect on β_2 -receptors at oral doses <100 mg (in adults); does not exhibit any membrane stabilizing or intrinsic sympathomimetic activity

Pharmacokinetics (Adult Data Unless Noted)

Note: The pharmacokinetics of metoprolol succinate ER capsule or tablet in hypertensive children and adolescents 6 to 17 years of age were found to be similar to adults.

Onset of action: Oral: IR tablets: Within 1 hour; Peak effect: Oral: 1 to 2 hours (Regardh 1980); IV: ~20 minutes (when infused over 10 minutes).

Duration: Oral: Immediate release: Variable (dose-related; 50% reduction in maximum heart rate after single doses of 20, 50, and 100 mg occurred at 3.3, 5, and 6.4 hours, respectively); Extended release: ~24 hours.

Absorption: Rapid and complete.

Distribution: V_d : 3.2 to 5.6 L/kg; crosses the blood brain barrier; CSF concentrations are 78% of plasma concentrations.

Protein binding: ~10% to 12% to albumin.

Metabolism: Extensively hepatic via CYP2D6; significant first-pass effect (~50%).

Bioavailability: Oral: Immediate release: ~40% to 50% (Johnsson 1975); Extended release: 77% relative to immediate release.

Half-life elimination: Neonates: 5 to 10 hours (Morselli 1989); Adults: 3 to 4 hours (7 to 9 hours in poor CYP2D6 metabolizers or hepatic impairment).

Excretion: Urine (95%, <5% to 10% as unchanged drug; increased to 30% to 40% in poor CYP2D6 metabolizers).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: Elimination half-life may be considerably prolonged, depending on severity.

Pharmacogenomics: Poor CYP2D6 metabolizers have several-fold higher metoprolol plasma concentrations.

Patient Counseling Points

(For additional information see "Metoprolol: Patient drug information")

What is this drug used for?

- It is used to treat high blood pressure.
- It is used to treat some types of chest pain (angina).
- It is used in certain people to lower the risk of death and having to go to the hospital because of heart failure.
- It is used after a heart attack to help prevent future heart attacks and lengthen life.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Feeling dizzy, tired, or weak

- Diarrhea, upset stomach, or throwing up
- Feeling cold in the arms or legs

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating
- Depression
- Severe dizziness or passing out
- Chest pain that is new or worse
- An abnormal heartbeat that is new or worse
- Slow heartbeat
- Shortness of breath, a big weight gain, or swelling in the arms or legs
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Apo metoprolol | Betaloc | Cadmet xr | Lopresor | Seloken xl;
- (AR) Argentina: Belozok | Lopresor;
- (AT) Austria: Beloc | Beloc duriles | Lanoc | Metohexal | Metomed | Metoprolol | Metoprolol actavis | Metoprolol genericon pharma | Metoprolol sandoz | Metoprolol stada | Metoprololtartrat Hexal | Metotens | Seloken;
- (AU) Australia: Apo metoprolol | Betaloc | Cm metoprolol | Genrx metoprolol | Lopresor | Metatar | Metohexal | Metolol | Metoprolol | Metoprolol actavis | Metoprolol an | Metoprolol mylan | Metoprolol rbx | Metoprolol sandoz | Metrol | Minax | Mistrom | Noumed metoprolol | Topreloc xl | Toprol xl | Tw metoprolol;
- (BD) Bangladesh: Angilat | Betacard | Betaloc | Betaone | Metaloc | Metocard | Metoprol xl | Preloc | Presonil | Prolol | Selomet | Topress;
- (BE) Belgium: Lopresor | Metoprolol EG | Metoprolol Mylan | Metoprolol sandoz | Metoprolol teva | Metoprolol teva generics belgium | Seloken | Selozok;

- (BF) Burkina Faso: Met xl;
- (BG) Bulgaria: Betablockade | Betaloc | Corvitol | Egilok | Lopresor | Mecorlong | Metocor | Metoprolol | Metoprolol tartrate accord | Metostad | Metostad cr | Promerol | Succiprol | Vaso-cardin;
- (BR) Brazil: Beca | Betacris | Dozoito | Emprol xr | Inephoros xr | Lopressor | Miclox | Minola | Quenzor | Seloken | Selozok | Succinato de metoprolol | Tartarato de metoprolol | Zarmine;
- (CH) Switzerland: Beloc | Lopresor | Meto Zerok | Metoprol zok mepha | Metoprolol Actavis Zerok | Metoprolol Axapharm | Metoprolol helvepharm | Metoprolol spirig hc | Metoprolol teva;
- (CI) Côte d'Ivoire: Met xl | Revelol-xl;
- (CL) Chile: Betaloc | Betaloc zok | Lopresor | Meprochem | Metoprolol tartrato | Rymsor;
- (CN) China: Betaloc | Heng de | Metoprolol | Shu meng | Yi jie er;
- (CO) Colombia: Beloc | Betaloc | Betaloc zok | Betoprolol | Carprol ls | Etoprol | Grandil | Lopresor | Metoprolol | Metoprolol mk | Metoprolol tartrato | Metpure xl | Presartin | Prezoc | Tensoprol;
- (CR) Costa Rica: Betaloc zok | Metoprolol tartrato | Metoproteg;
- (CZ) Czech Republic: Apo metoprolol | Betaloc | Betaloc zok | Bloxazoc | Egilok | Egilok succ | Emzok | Hypablo | Metoprolol | Metoprolol al | Metoprolol aurovitas | Metoprolol dh | Metoprolol medreg | Metoprolol Mylan | Vasocardin;
- (DE) Germany: Agoloc | Azumetop | Beloc | Beloc duriles | Beloc zok | Beloc zok mite | Betaloc zok | Dignometoprol | Jeprolol | Jutabloc | Lopresor | Lopressor | Meprolo | Metizok | Meto | Meto biochemie | Meto Hennig | Meto isis | Meto phamos | Meto tablinen | Meto-isis NT | Meto-Puren | Metoberag | Metobeta | Metodoc | Metodura | Metodura znt | Metohexal | Metohexal Succ | Metok | Metomerck | Metoprogamma | Metoprogamma succinat | Metoprolol | Metoprolol - 1 a pharma | Metoprolol 1 A Pharma Succ | Metoprolol abz | Metoprolol actavis | Metoprolol al | Metoprolol axcount | Metoprolol bioeq pharma | Metoprolol carino | Metoprolol carinopharm | Metoprolol Corax | Metoprolol CT | Metoprolol ct zero | Metoprolol Gry | Metoprolol Heumann | Metoprolol Heumann Heunet | Metoprolol ratiopharm Succinat | Metoprolol sandoz | Metoprolol stada | Metoprolol Succ CT | Metoprolol succinat acino | Metoprolol succinat actavis | Metoprolol succinat Beta | Metoprolol Succinat tad | Metoprolol Verla | Metoprolol von ct | Metoprolol wolff | Metoprolol z al | Metoprolol ZOT stada | Metoprolol-ratiopharm | Metoprololsuccinat 1 A Pharma | Metoprololsuccinat aaa | Metoprololsuccinat AbZ | Metoprololsuccinat AL | Metoprolol-succinat bioeq pharma | Metoprololsuccinat dura | Metoprololsuccinat heumann | Metoprololsuccinat STADA | Metoprololsuccinat amneal | Metoprololsuccinate Teva | Metoprololsuccinate Winthrop | Metowieb | Prelis | Selokeen | Seloken | Selozok | Sigaprolol;
- (DK) Denmark: Metocar | Metoprolol GEA | Metoprololsuccinat 1a farma | Metoprololsuccinat hexal | Metoratio;
- (DO) Dominican Republic: Betaloc zok | Cardeler | Lopresor | Meprolo | Metopres | Metoprolol | Metoprolol lam | Prolaken | Ritmol | Tadoril | Vidacor;
- (EC) Ecuador: Betaloc zok | Metoprolol | Metoprolol mk;
- (EE) Estonia: Betaloc | Betaloc zok | Bloxazoc | Corvitol | Emzok | Lopresor | Metohexal | Metoprolol | Metoprolol Polpharma | Metoprolol ratiopharm | Metoprolol stada | Metoprolol-ratiopharm | Metozok | Spesicor | Spesimax | Vasocardin;
- (EG) Egypt: Betaloc | Carditag xl | Lopresor | Lowpress;
- (ES) Spain: Beloken | Lopresor | Metoprolol aurovitas | Metoprolol krka;
- (ET) Ethiopia: Betaloc zok | Melopres | Metoprolol;
- (FI) Finland: Betaloc | Metblock | Metohexal | Metomylan | Metopocor | Metoprolin | Metoprolol krka | Metoprolol orion | Metoprolol ratiopharm | Metoprolol tartrate hexal | Seloken | Seloken zoc | Selopral | Spesicor | Spesimax;
- (FR) France: Lopressor | Metoprolol accord | Metoprolol actavis | Metoprolol arrow | Metoprolol g gam | Metoprolol Ranbaxy | Metoprolol rpg | Metoprolol sandoz | Seloken | Selozok | Selozok LP;
- (GB) United Kingdom: Arbralene | Betaloc | Lopresor | Mepranix | Metoprolol | Metoprolol berk | Metoprolol cox | Metoprolol kent | Metoprolol sandoz | Tensomex | Toprol xl;
- (GR) Greece: Beloc | Lopresor | Metazero;
- (GT) Guatemala: Betaloc zok;
- (HK) Hong Kong: Apo metoprolol | Betaloc | Betaloc zok | Metolol cp | Metoprolol | Metoprolol krka | Metracin | Minax | Slow metoprolol | Syn-Metoprolol;
- (HR) Croatia: Betaloc | Betaloc zok | Bloxan;

- (HU) Hungary: Betaloc | Egilok | Huma-metoprol | Metohexal | Metoprolol al | Metoprolol onkogen | Metoprolol-b | Metoprolol-ratiopharm | Ritmetol;
- (ID) Indonesia: Betaloc zok | Cardiosel | Fapresor | Lopressor | Loprolol | Seloken | Selozok;
- (IE) Ireland: Betaloc | Betazok | Lopresor | Metocor | Metop;
- (IL) Israel: Lopresor | Metopress | Neobloc;
- (IN) India: Acmeto xl | Aimet xr | Almetolol | Ambimet xl | Angibloc ER | Arelol 25 sr | Avelprol | Beboxia 25 xl | Beboxia 50 xl | Belol | Belol xl | Best beta | Betabest | Betafit | Betaloc | Betamet xl | Betaone | Betaone xl | Blumeta | Cardibeta | Cymet | Dilnorm | Embeta | Emlol | Evimeto | Ibmeto xr | Kplol xl | Lakmet | Libmet xl | Lopresor | Mataprol | Megapralol xl | Meptolol | Met xl | Metacure | Metaloc | Metapro | Metapro-XL | Metcy | Metfine xl | Metfirst xl | Metheart xl | Metis er | Meto | Metoan er | Metocard xl | Metodex XL | Metodil xl | Metograf | Metogud xl | Metol xl | Metolar | Metolar XR | Metonce | Metonex | Metopil | Metopot | Metoprol xl | Metosan xr | Metox | Metozaar | Metozox | Metprocer xl | Metproleap | Metpure | Metpure xl | Metsave xl | Metscore | Metspot | Metsun xl | Metvio cr | Metvolen xl | Metzok | Mexcelol | Mexmet | Mpl | Mprol | Mt loc | Mymeto xl | Myoprol XL | Naypres xl | Normomet xl | Numet | Oldbeta xl | Ozolol xl | Prolomet xl | Prolovas xl | Protol | Restopress | Revelol | Revelol-xl | Seloken | Sitelol | Sms-XL | Starpress XL | Supermet xl | Sustamet | Sustameto | Tamepro | Tecbeta | Translol xl | Trulol | Velol xl | Vinicor-XL | Vitameto xr | Vivalol | Voizmet | Xm Beta | Yogaprol | Zenmet xl;
- (IQ) Iraq: Meto rx | Metoprolol awa | Metoprolol kindi | Metoral | Metorx;
- (IT) Italy: Betaloc zok | Lopresor | Metoprololo | Metoprololo aristo | Metoprololo aurobindo | Metoprololo Doc | Metoprololo eg | Metoprololo hexal | Metoprololo krka | Seloken;
- (JO) Jordan: Betaloc | Lopresor;
- (JP) Japan: Cerekunart | Hanilost | Kalsuhon | Kokonalin | Lopresor | Lopresor sr | Medepine | Melucomon | Metolastar | Metoplic | Metoplic taiyo | Metoprolol tartrate sawai | Metoprolol tartrate towa | Mitacard | Selektion | Selektion choseido | Seloken | Seloken I | Shinzanol amel | Sioptrol | Sipseron | Zegmular | Zizel;
- (KE) Kenya: Betaloc zok | Lopressor | Met xl | Metpure xl | Revelol-xl;
- (KR) Korea, Republic of: Betaloc | Prolol;
- (KW) Kuwait: Betaloc | Lopresor;
- (LB) Lebanon: Betablok sdk | Betaloc zok | Lopresor;
- (LT) Lithuania: Apo metoprolol | Betaloc | Bloxan | Corvitol | Emzok | Lopressor | Metazero | Meto isis | Meto-Puren | Metocard | Metohexal | Metohexal Succ | Metoprolol | Metoprolol Polpharma | Metoprolong | Spesicor | Vasocardin;
- (LU) Luxembourg: Lopresor | Metohexal | Metohexal Succ | Metoprolol EG | Prelis | Seloken | Selozok;
- (LV) Latvia: Apo metoprolol | Betaloc | Betaloc zok | Bloxan | Corvitol | Egilok | Emzok | Lopresor | Metazero | Meto | Metohexal | Metoprogamma | Metoprolol | Metoprolol hexal z | Metoprolol Meda | Metoprolol Polpharma | Metoprolol ratiopharm | Spesicor | Vasocardin;
- (MA) Morocco: Lopressor;
- (MX) Mexico: Apoprole | Betaloc zok | Bioprol | Delandex | Eurolool | Futaline | Kapirmetol | Kenaprol | Laera | Lenopres | Lopresor | Lopresor r | Lopritol | Metobest | Metopresol | Metoprolol | Metoprolol avivia | Metoprolol g.i | Metoprolol jayor | Metoprolol kener | Metoprolol tecnofarma | Metoprolol victory | Mezelol | Nipresol | Ovaxtyl | Proken-m | Prolaken | Promiced | Prontol | Ritmolol | Ruteral | Seloken | Seloken zok | Sermetrol | Synadrenol | Tiazidol;
- (MY) Malaysia: Aft metoprolol cr | Apo metoprolol | Beatrolol | Betaloc | Betatab | Betawin | Denex | Hypersor | Metohexal | Metoprolol | Metoprolol Pharmaniaga;
- (NG) Nigeria: Betaloc zok | Met xl | Metoloc xl | Metpure xl;
- (NL) Netherlands: Lopresor | Metoprolol | Metoprololsuccinaat 1A pharma retard | Metoprololsuccinaat ratiopharm | Metoprololsuccinaat sandoz | Metoprololtartraat Accord | Metoprololtartraat actavis | Metoprololtartraat Alpharma | Metoprololtartraat aurobindo | Metoprololtartraat CF | Metoprololtartraat flx | Metoprololtartraat Mylan | Metoprololtartraat pch | Metoprololtartraat ratiopharm | Metoprololtartraat Sandoz | Metoprololtartraat A | Selokeen | Selokeen zoc | Seloken | Selozok;
- (NO) Norway: Betaloc zok | Bloxazoc | Metopocor | Metoprolol | Metoprolol 50 1a pharma | Metoprolol actavis | Metoprolol al 50 nordic prime | Metoprolol arrow lab | Metoprolol mylan | Metoprolol retard gea | Metoprolol retard sandoz | Metoprolol sandoz | Metoprolol teva |

- Metoprolol vp bausch | Metoprololo aurobindo | Metoprololo eg | Metoprololtartrat nordic prime | Selo zok | Seloken;
- (NZ) New Zealand: Apo metoprolol | Betaloc | Ipca metoprolol | Lopresor | Metoprolol iv viatris | Metoprolol mylan;
 - (PA) Panama: Betaloc zok;
 - (PE) Peru: Betaloc | Betaloc zok | Embeta xr | Revelol-xl;
 - (PH) Philippines: Angimet | Angiobloc | Angionorm | Betaloc | Betaryx | Betavasc | Betazok | Boie Metoprolol Tartrate | Cardiopin | Cardiosel | Cardiostat | Corbloc | Embeta xr | Gerbloc | Hypetor | Loprexo | Metobloc | Metocard | Metocare | Metoprim | Metoprolol | Metoprolol Interhealth Care | Metospec | Metostad | Montebloc | Neobloc | Neox | Neox ER | Princicar | Prolat | Prolohex | Prolol | Prometin | Recebloc | Sanerg | Valvexin | Zionel;
 - (PK) Pakistan: Aksopresor | Betaloc | Carsel | Liskoprol | Lopresor | Mepresor | Meprol | Merol | Metapressor | Metocard | Metopro | Metsim | Refit;
 - (PL) Poland: Beloc | Betaloc | Betaloc zok | Beto 100 zk | Beto 150 zk | Beto 200 zk | Beto 25 zk | Beto 50 zk | Bloxazoc | Iviprolol | Metocard | Metocard zk | Metogen sc | Metohexal | Metoprolol | Metoprolol 1A Pharma | Metoprolol biofarm | Metoprolol medreg | Metoprololum 123ratio | Metoratio Zk | Metostad | Selmet | Seloken | Symlok;
 - (PR) Puerto Rico: Kapsargo sprinkle | Lopresor | Metoprolol | Toprol xl;
 - (PT) Portugal: Lopresor | Metoprolol | Metoprolol Aurobindo;
 - (PY) Paraguay: Lopresor | Metpure xl;
 - (QA) Qatar: Apo-Metoprolol | Betaloc | Lopresor | Metloc | Metolar;
 - (RO) Romania: Betaloc | Betaloc zok | Betaprol | Bloxan | Egilok | Egilok ep | Emzok | Metoprolol medreg | Metoprolol zentiva | Metosuccinat | Vasocardin;
 - (RU) Russian Federation: Betaloc | Betaloc zok | Betalok | Corvitol | Egilok | Emzok | Metocard | Metocor | Metolol | Metoprolol | Metoprolol acry | Metoprolol akri | Metoprolol Akrikhin | Metoprolol krka | Metoprolol obl | Metoprolol Obolensk | Metoprolol organika | Metoprolol retard akrikhin | Metoprolol teva | Metoprolol velfarm | Metoprolol vertex | Metoprolol zentiva | Metoprolol-ratiopharm | Metozok | Serdol | Vasocardin;
 - (SA) Saudi Arabia: Apo metoprolol | Carelio | Lopresor | Metracin | Pms-metoprolol | Pms-metoprolol-l;
 - (SE) Sweden: Bloxazoc | Metomylan | Metoproferm | Metoprolol 1A Farma Retard | Metoprolol actavis | Metoprolol biochemie | Metoprolol ebb | Metoprolol GEA | Metoprolol Hexal | Metoprolol ipex | Metoprolol Orifarm | Metoprolol orion | Metoprolol ratiopharm | Metoprolol sandoz | Mozoc | Seloken | SelokenZoc;
 - (SG) Singapore: Betaloc | Denex;
 - (SI) Slovenia: Beloc | Bloxan | Metoprolol carinopharm | Seloken;
 - (SK) Slovakia: Betaloc | Corvitol | Egilok | Egilok xl | Metoprolol medreg | Metoprolol Mylan | Metoprolol ratiopharm | Vasocardin;
 - (SR) Suriname: Lopresor | Metolar | Metoprololtartraat | Metoprololtartraat aurobindo | Metoprololtartraat ratiopharm;
 - (SY) Syrian Arab Republic: Hypopresor er;
 - (TH) Thailand: Betaloc | Betaloc zok | Cardeloc | Cardoxone r | Denex | Hyperloc | Lopresor | Meloc | Melol | Metoblock | Metocard | Metoloc | Metolol | Metoprolol | Metprolol | Minax | Sefloc;
 - (TN) Tunisia: Lopresor | Lopresor LP | Vasocard LP | Vasocardin;
 - (TR) Turkey: Beloc | Beloc zok | Betablok sdk | Cardovol | Lopresor | Mepolex | Metloc | Probloc | Problok | Saneloc;
 - (TW) Taiwan: Apo metoprolol | Betaloc | Betapress | Betterlock | Cancliol | Cartaloc | Denex;
 - (UA) Ukraine: Betaloc | Corvitol | Egilok | Egilok retard | Emzok | Metoprol retard | Metoprolol | Metoprolol Tartras | Metoprolol xl | Metoprolol zentiva | Vasocardin;
 - (UG) Uganda: Betaloc zok | Met xl | Metomac | Metpure xl | Miopolol er;
 - (UY) Uruguay: Belozok | Lopresor | Mebuten | Zion;
 - (VE) Venezuela, Bolivarian Republic of: Betaloc | Lopresor | Metoprolol;
 - (ZA) South Africa: Lopresor;
 - (ZM) Zambia: Metpure xl | Revelol-xl;
 - (ZW) Zimbabwe: Betaloc zok | Revelol-xl

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REFERENCES

1. Acikel S, Bozbas H, Gultekin B, et al. Comparison of the efficacy of metoprolol and carvedilol for preventing atrial fibrillation after coronary bypass surgery. *Int J Cardiol.* 2008;126(1):108-113. [PubMed 17499863]
2. Ahlfors CE. Benzyl alcohol, kernicterus, and unbound bilirubin. *J Pediatr.* 2001;139(2):317-319. doi:10.1067/mpd.2001.116281 [PubMed 11487763]
3. Ahmed AI, van Mierlo P, Jansen P. Sleep disorders, nightmares, depression and anxiety in an elderly patient treated with low-dose metoprolol. *Gen Hosp Psychiatry.* 2010;32(6). doi:10.1016/j.genhosppsych.2010.04.008 [PubMed 21112460]
4. Ailani J, Burch RC, Robbins MS; Board of Directors of the American Headache Society. The American Headache Society consensus statement: Update on integrating new migraine treatments into clinical practice. *Headache.* 2021;61(7):1021-1039. doi:10.1111/head.14153 [PubMed 34160823]
5. Al-Khatib SM, Stevenson WG, Ackerman MJ, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society [published online October 30, 2017]. *Circulation.* doi:10.1161/CIR.0000000000000549 [PubMed 29084731]
6. Allen LV Jr, Erickson MA 3rd. Stability of labetalol hydrochloride, metoprolol tartrate, verapamil hydrochloride, and spironolactone with hydrochlorothiazide in extemporaneously compounded oral liquids. *Am J Health Syst Pharm.* 1996;53(19):2304-2309. [PubMed 8893069]
7. American College of Obstetricians and Gynecologists (ACOG). ACOG Practice Bulletin No. 203: Chronic hypertension in pregnancy. *Obstet Gynecol.* 2019;133(1):e26-e50. [PubMed 30575676]
8. American College of Obstetricians and Gynecologists (ACOG). Headaches in pregnancy and postpartum: ACOG clinical practice guideline No. 3. *Obstet Gynecol.* 2022;139(5):944-972. doi:10.1097/AOG.0000000000004766 [PubMed 35576364]
9. American Society of Health-System Pharmacists (ASHP). Standardize 4 safety. <https://www.ashp.org/pharmapactice/standardize-4-safety-initiative>. Updated March 2025. Accessed February 19, 2026.
10. Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther.* 2016;100(1):42-52. doi:10.1089/bfm.2017.0077 [PubMed 27060684]
11. Aronow WS, Fleg JL, Pepine CJ, et al; ACCF Task Force. ACCF/AHA 2011 expert consensus document on hypertension in the elderly: a report of the American College of Cardiology Foundation Task Force on Clinical Expert Consensus Documents. *Circulation.* 2011;123(21):2434-2506. [PubMed 21518977]
12. Bahn RS, Burch HB, Cooper DS, et al. Hyperthyroidism and other causes of thyrotoxicosis: management guidelines of the American Thyroid Association and American Association of Clinical Endocrinologists. *Endocr Pract.* 2011;17(3):456-520. [PubMed 21700562]
13. Bankhead R, Boullata J, Brantley S, et al. Enteral nutrition practice recommendations. *JPEN J Parenter Enteral Nutr.* 2009;33(2):122-167. doi:10.1177/0148607108330314 [PubMed 19171692]
14. Barron AJ, Zaman N, Cole GD, Wensel R, Okonko DO, Francis DP. Systematic review of genuine versus spurious side-effects of beta-blockers in heart failure using placebo control: recommendations for patient information. *Int J Cardiol.* 2013;168(4):3572-3579. doi:10.1016/j.ijcard.2013.05.068 [PubMed 23796325]
15. Blaszczyk A, Brandt N, Ashley J, Tuders N, Doles H, Stefanacci RG. Crushed tablet administration for patients with dysphagia and enteral feeding: challenges and considerations. *Drugs Aging.* 2023;40(10):895-907. doi:10.1007/s40266-023-01056-y [PubMed 37707775]
16. Boullata JI. Drug administration through an enteral feeding tube. *Am J Nurs.* 2009;109(10):34-42. doi:10.1097/01.NAJ.0000361488.45094.28 [PubMed 21753476]

17. Boullata JI. *Guidebook on Enteral Medication Administration*. American Society for Parenteral and Enteral Nutrition; 2019.
18. Boullata JI, Carrera AL, Harvey L, et al. ASPEN safe practices for enteral nutrition therapy. *JPEN J Parenter Enteral Nutr*. 2017;41(1):15-103. doi:10.1177/0148607116673053 [PubMed 27815525]
19. Bratt EL, Östman-Smith I. Effects of lifestyle changes and high-dose β -blocker therapy on exercise capacity in children, adolescents, and young adults with hypertrophic cardiomyopathy. *Cardiol Young*. 2015;25(3):501-510. [PubMed 24607033]
20. Burnett AL, Nehra A, Breau RH, et al. Erectile dysfunction: AUA guideline. *J Urol*. 2018;200(3):633-641. doi:10.1016/j.juro.2018.05.004 [PubMed 29746858]
21. Burkauskas J, Noreikaite A, Bunevicius A, et al. Beta-1-selective beta-blockers and cognitive functions in patients with coronary artery disease: a cross-sectional study. *J Neuropsychiatry Clin Neurosci*. 2016;28(2):143-146. doi:10.1176/appi.neuropsych.15040088 [PubMed 26715033]
22. Centers for Disease Control (CDC). Neonatal deaths associated with use of benzyl alcohol—United States. *MMWR Morb Mortal Wkly Rep*. 1982;31(22):290-291. [PubMed 6810084]
23. Chatzidou S, Kontogiannis C, Tsilimigras DI, et al. Propranolol versus metoprolol for treatment of electrical storm in patients with implantable cardioverter-defibrillator. *J Am Coll Cardiol*. 2018;71(17):1897-1906. doi:10.1016/j.jacc.2018.02.056 [PubMed 29699616]
24. Chaudhary I. Microvascular angina: Angina pectoris due to coronary microvascular dysfunction. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 9, 2021.
25. Chen ZM, Pan HC, Chen YP, et al; COMMIT (CLOpidogrel and Metoprolol in Myocardial Infarction Trial) collaborative group. Early intravenous then oral metoprolol in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet*. 2005;366(9497):1622-1632. [PubMed 16271643]
26. Cífková R, Johnson MR, Kahan T, et al. Peripartum management of hypertension: a position paper of the ESC Council on Hypertension and the European Society of Hypertension. *Eur Heart J Cardiovasc Pharmacother*. 2020;6(6):384-393. doi:10.1093/ehjcvp/pvz082 [PubMed 31841131]
27. Cocco G. Erectile dysfunction after therapy with metoprolol: the Hawthorne effect. *Cardiology*. 2009;112(3):174-177. doi:10.1159/000147951 [PubMed 18654082]
28. Covar RA, Macomber BA, Szeffler SJ. Medications as asthma triggers. *Immunol Allergy Clin North Am*. 2005;25(1):169-190. doi:10.1016/j.iac.2004.09.009 [PubMed 15579370]
29. Cove-Smith JR, Kirk CA. CNS-related side-effects with metoprolol and atenolol. *Eur J Clin Pharmacol*. 1985;(28)(suppl):69-72. doi:10.1007/BF00543713 [PubMed 4054193]
30. Croft CH, Rude RE, Gustafson N, et al. Abrupt withdrawal of beta-blockade therapy in patients with myocardial infarction: effects on infarct size, left ventricular function, and hospital course. *Circulation*. 1986;73(6):1281-1290. doi:10.1161/01.cir.73.6.1281 [PubMed 3009050]
31. Dahlöf C, Dimenäs E. Side effects of beta-blocker treatments as related to the central nervous system. *Am J Med Sci*. 1990;299(4):236-244. doi:10.1097/00000441-199004000-00004 [PubMed 1969714]
32. Day JL, Metcalfe J, Simpson CN. Adrenergic mechanisms in control of plasma lipid concentrations. *Br Med J (Clin Res Ed)*. 1982;284(6323):1145-1148. doi:10.1136/bmj.284.6323.1145 [PubMed 6122483]
33. Diener HC, Hartung E, Chrubasik J, et al; Study Group. A comparative study of oral acetylsalicylic acid and metoprolol for the prophylactic treatment of migraine. A randomized, con-

- trolled, double-blind, parallel group phase III study. *Cephalalgia*. 2001;21(2):120-128. [PubMed 11422094]
34. Doyon JB, Liu KJ, Berman RA. Metoprolol-induced total body erythroderma. *J Gen Intern Med*. 2017;32(2):221-222. doi:10.1007/s11606-016-3900-2 [PubMed 27761769]
 35. Dransfield MT, Voelker H, Bhatt SP, et al. Metoprolol for the Prevention of Acute Exacerbations of COPD. *N Engl J Med*. 2019;381(24):2304-2314. doi:10.1056/NEJMoa1908142 [PubMed 31633896]
 36. Drayer DE. Lipophilicity, hydrophilicity, and the central nervous system side effects of beta blockers. *Pharmacotherapy*. 1987;7(4):87-91. doi:10.1002/j.1875-9114.1987.tb04029.x [PubMed 2891122]
 37. Dungan K, Merrill J, Long C, Binkley P. Effect of beta blocker use and type on hypoglycemia risk among hospitalized insulin requiring patients. *Cardiovasc Diabetol*. 2019;18(1):163. Published 2019 Nov 27. doi:10.1186/s12933-019-0967-1 [PubMed 31775749]
 38. Duthaler U, Bachmann F, Suenderhauf C, et al. Liver cirrhosis affects the pharmacokinetics of the six substrates of the basal phenotyping cocktail differently. *Clin Pharmacokinet*. 2022;61(7):1039-1055. doi:10.1007/s40262-022-01119-0 [PubMed 35570253]
 39. Dvořáčková E, Pilková A, Matoulek M, Slanař O, Hartinger JM. Bioavailability of orally administered drugs after bariatric surgery. *Curr Obes Rep*. 2024;13(1):141-153. doi:10.1007/s13679-023-00548-7 [PubMed 38172482]
 40. Effect of metoprolol CR/XL in chronic heart failure: Metoprolol CR/XL Randomised Intervention Trial in Congestive Heart Failure (MERIT-HF). *Lancet*. 1999;353(9169):2001-2007. [PubMed 10376614]
 41. Egstrup K. Transient myocardial ischemia after abrupt withdrawal of antianginal therapy in chronic stable angina. *Am J Cardiol*. 1988;61(15):1219-1222. doi:10.1016/0002-9149(88)91158-7 [PubMed 2897781]
 42. Expert opinion. Senior Enteral Feeding Tube Editorial Team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.
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 45. Farmakis IT, Pyrgidis N, Doundoulakis I, Mykoniatis I, Akrivos E, Giannakoulas G. Effects of major antihypertensive drug classes on erectile function: a network meta-analysis. *Cardiovasc Drugs Ther*. Published online May 4, 2021. doi:10.1007/s10557-021-07197-9 [PubMed 33945044]
 46. Field JM, Hazinski MF, Sayre MR, et al. Part 1: Executive Summary: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation*. 2010;122(18)(suppl 3):640-656. [PubMed 20956217]
 47. Fisher AA, Davis M, Jeffery I. Acute delirium induced by metoprolol. *Cardiovasc Drugs Ther*. 2002;16(2):161-165. doi:10.1023/a:1015761618314 [PubMed 12090909]
 48. Flynn JT, Kaelber DC, Baker-Smith CM, et al; Subcommittee on Screening and Management of High Blood Pressure in Children. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. *Pediatrics*. 2017;140(3):e20171904. [PubMed 29192011]
 49. Fonarow GC, Abraham WT, Albert NM, et al. Influence of beta-blocker continuation or withdrawal on outcomes in patients hospitalized with heart failure: findings from the OPTIMIZE-HF program. *J Am Coll Cardiol*. 2008;52(3):190-199. doi:10.1016/j.jacc.2008.03.048 [PubMed 18617067]

50. Fonseca VA. Effects of beta-blockers on glucose and lipid metabolism. *Curr Med Res Opin.* 2010;26(3):615-629. doi:10.1185/03007990903533681 [PubMed 20067434]
51. Foster CA, Aston SJ. Propranolol-Epinephrine Interaction: A Potential Disaster. *Plast Reconstr Surg.* 1983;72(1):74-78. [PubMed 6867180]
52. Frishman WH. Beta-adrenergic receptor blockers. Adverse effects and drug interactions. *Hypertension.* 1988;11(3, pt 2):II21-II29. doi:10.1161/01.hyp.11.3.pt_2.ii21 [PubMed 2895072]
53. Frishman WH. Beta-adrenergic blocker withdrawal. *Am J Cardiol.* 1987;59(13):26F-32F. doi:10.1016/0002-9149(87)90038-5 [PubMed 2883874]
54. Gesquiere I, Darwich AS, Van der Schueren B, et al. Drug disposition and modelling before and after gastric bypass: immediate and controlled-release metoprolol formulations. *Br J Clin Pharmacol.* 2015;80(5):1021-1030. doi:10.1111/bcp.12666 [PubMed 25917170]
55. Gleiter CH, Deckert J. Adverse CNS-effects of beta-adrenoceptor blockers. *Pharmacopsychiatry.* 1996;29(6):201-211. doi:10.1055/s-2007-979572 [PubMed 8956349]
56. Grundmann M, Kacirova I, Duricova, J et al. "Metoprolol and alfa-hydroxymetoprolol concentrations during lactation-a case report." 12th International Congress of Therapeutic Drug Monitoring and Clinical Toxicology Stuttgart, Germany · October 2 – 6, 2011. *Ther Drug Monit.* 2011;33(4):504.
57. Haghighi M, Saravi M, Hashemi MJ, et al. Optimal beta-blocker for prevention of atrial fibrillation after on-pump coronary artery bypass graft surgery: carvedilol versus metoprolol. *Heart Rhythm.* 2007;4(9):1170-1174. [PubMed 17765616]
58. Hansen T, Fynne L. Jaundice and liver injury with cholestatic pattern after treatment with metoprololsuccinat [in Danish]. *Ugeskr Laeger.* 2017;179(51):V09170698. [PubMed 29260695]
59. Harrison DC, Alderman EL. Editorial: Discontinuation of propranolol therapy. Cause of rebound angina pectoris and acute coronary events. *Chest.* 1976;69(1):1-2. doi:10.1378/chest.69.1.1 [PubMed 1244260]
60. Hawkins DN, Faler BJ, Choi YU, Prasad BM. Time course of blood pressure decrease after bariatric surgery in normotensive and hypertensive patients. *Obes Surg.* 2018;28(7):1845-1851. doi:10.1007/s11695-017-3091-x [PubMed 29725978]
61. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2022;145(18):e895-e1032. doi:10.1161/CIR.0000000000001063 [PubMed 35363499]
62. Hillis LD, Smith PK, Anderson JL, et al. 2011 ACCF/AHA guideline for coronary artery bypass graft surgery: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation.* 2011;124(23):2610-2642. [PubMed 22064600]
63. Hirsch AT, Haskal ZJ, Hertzner NR, et al. ACC/AHA Guidelines for the Management of Patients With Peripheral Arterial Disease (Lower Extremity, Renal, Mesenteric, and Abdominal Aortic): Executive Summary. A Collaborative Report of the American Association for Vascular Surgery/Society for Vascular Surgery, Society for Cardiovascular Angiography and Interventions, Society for Vascular Medicine and Biology, Society of Interventional Radiology, and the ACC/AHA Task Force on Practice Guidelines (Writing Committee to Develop Guidelines for the Management of Patients With Peripheral Arterial Disease). *Circulation.* 2006;113(11):1474-1547. [PubMed 16549646]
64. Ho TK, Moretti ME, Schaeffer JK et al. Maternal beta-blocker usage and breast feeding in the neonate. The American Pediatric Society and The Society for Pediatric Research 1999 annual meeting. San Francisco, California, USA. May 1-4, 1999. Abstracts. *Pediatr Res.* 1999;45(4, pt 2):67A. [PubMed 10372206]
65. Hoffmann KJ, Regårdh CG, Aurell M, Ervik M, Jordö L. The effect of impaired renal function on the plasma concentration and urinary excretion of metoprolol metabolites. *Clin Pharmacokinet.* 1980;5(2):181-191. doi:10.2165/00003088-198005020-00005 [PubMed 7363533]

66. Huckleberry Y, Thomas MC, Erstad BL. Dosage conversions as a potential cause of adverse drug events. *Am J Health Syst Pharm*. 2003;60(2):189-191. [PubMed 12561664]
67. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics (AAP) Committee on Drugs. *Pediatrics*. 1997;99(2):268-278. doi:10.1542/peds.99.2.268 [PubMed 9024461]
68. Ito S. Drug therapy for breast-feeding women. *N Engl J Med*. 2000;343(2):118-126. [PubMed 10891521]
69. Jaillon P. Relevance of intrinsic sympathomimetic activity for beta blockers. *Am J Cardiol*. 1990;66(9):21C-23C. doi:10.1016/0002-9149(90)90758-s [PubMed 1977302]
70. James PA, Oparil S, Carter BL, et al. 2014 evidence-based guideline for the management of high blood pressure in adults: report from the panel members appointed to the Eighth Joint National Committee (JNC 8). *JAMA*. 2014;311(5):507-520. [PubMed 24352797]
71. Joglar JA, Chung MK, Armbruster AL, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Circulation*. 2024;149(1):e1-e156. doi:10.1161/CIR.0000000000001193 [PubMed 38033089]
72. Johnsson G, Regårdh CG, Sölvell L. Combined pharmacokinetic and pharmacodynamic studies in man of the adrenergic beta1-receptor antagonist metoprolol. *Acta Pharmacol Toxicol (Copenh)*. 1975;36(suppl 5):31-44. [PubMed 1094802]
73. Jones DW, Ferdinand KC, Taler SJ, et al. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PC guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152(11):e114-e218. doi:10.1161/CIR.0000000000001356 [PubMed 40811497]
74. Jordö L, Attman PO, Aurell M, Johansson L, Johnsson G, Regårdh CG. Pharmacokinetic and pharmacodynamic properties of metoprolol in patients with impaired renal function. *Clin Pharmacokinet*. 1980;5(2):169-180. doi:10.2165/00003088-198005020-00004 [PubMed 6102500]
75. Kaplan R. Atrial fibrillation in adults: Rate control in patients without heart failure. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed February 28, 2024.
76. Kapsargo (metoprolol) [prescribing information]. Cranbury, NJ: Sun Pharmaceuticals; May 2018.
77. Kapsargo Sprinkle (metoprolol) [prescribing information]. New Brunswick, NJ: Ohm Laboratories Inc; April 2023.
78. Kettinger K, Mewis C, Dörnberger V, Vonthein R, Bosch RF, Kühlkamp V. Efficacy of metoprolol and sotalol in the prevention of recurrences of sustained ventricular tachyarrhythmias in patients with an implantable cardioverter defibrillator. *Pacing Clin Electrophysiol*. 2002;25(11):1571-1576. [PubMed 12494613]
79. Kim Y, Miller M. Marked hypertriglyceridemia in a woman receiving metoprolol succinate. *J Clin Lipidol*. 2014;8(6):640-643. doi:10.1016/j.jacl.2014.07.009 [PubMed 25499948]
80. Kir F, Al-Sulaiti FK, Sahin S. Evaluation of in vitro dissolution profiles of modified-release metoprolol succinate tablets crushed using mortar and pestle technique. *Eur J Pharm Sci*. 2024;194:106694. doi:10.1016/j.ejps.2024.106694 [PubMed 38191064]
81. Klang MG. Developing guidance for feeding tube administration of oral medications. *JPEN J Parenter Enteral Nutr*. 2023;47(4):519-540. doi:10.1002/jpen.2490 [PubMed 36847617]
82. Kleinbaum J, Shamoon H. Effect of propranolol on delayed glucose recovery after insulin-induced hypoglycemia in normal and diabetic subjects. *Diabetes Care*. 1984;7(2):155-162. doi:10.2337/diacare.7.2.155 [PubMed 6376011]

83. Kliegman RM, Stanton BF, St. Gemell JW, et al, eds. *Nelson Textbook of Pediatrics*. 19th ed. Saunders Elsevier; 2011.
84. Ko DT, Hebert PR, Coffey CS, Sedrakyan A, Curtis JP, Krumholz HM. Beta-blocker therapy and symptoms of depression, fatigue, and sexual dysfunction. *JAMA*. 2002;288(3):351-357. doi:10.1001/jama.288.3.351 [PubMed 12117400]
85. Koella WP. CNS-related (side-)effects of beta-blockers with special reference to mechanisms of action. *Eur J Clin Pharmacol*. 1985;28 Suppl:55-63. doi:10.1007/BF00543711 [PubMed 2865151]
86. Kostis JB, Rosen RC. Central nervous system effects of beta-adrenergic-blocking drugs: the role of ancillary properties. *Circulation*. 1987;75(1):204-212. doi:10.1161/01.cir.75.1.204 [PubMed 2878741]
87. Kotlyar E, Keogh AM, Macdonald PS, Arnold RH, McCaffrey DJ, Glanville AR. Tolerability of carvedilol in patients with heart failure and concomitant chronic obstructive pulmonary disease or asthma. *J Heart Lung Transplant*. 2002;21(12):1290-1295. doi:10.1016/s1053-2498(02)00459-x [PubMed 12490274]
88. Kuck KH, Cappato R, Siebels J, Rüppel R. Randomized comparison of antiarrhythmic drug therapy with implantable defibrillators in patients resuscitated from cardiac arrest: the Cardiac Arrest Study Hamburg (CASH). *Circulation*. 2000;102(7):748-754. [PubMed 10942742]
89. Kusumoto FM, Schoenfeld MH, Barrett C, et al. 2018 ACC/AHA/HRS guideline on the evaluation and management of patients with bradycardia and cardiac conduction delay: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines, and the Heart Rhythm Society. *J Am Coll Cardiol*. 2019;74(7):932-987. doi:10.1016/j.jacc.2018.10.043 [PubMed 30412710]
90. Lager I, Blohmé G, Smith U. Effect of cardioselective and non-selective beta-blockade on the hypoglycaemic response in insulin-dependent diabetics. *Lancet*. 1979;1(8114):458-462. doi:10.1016/s0140-6736(79)90821-3 [PubMed 85053]
91. Lammers JW, Folgering HT, van Herwaarden CL. Respiratory tolerance of bisoprolol and metoprolol in asthmatic patients. *J Cardiovasc Pharmacol*. 1986;8 Suppl 11:S69-S73. doi:10.1097/00005344-198511001-00012 [PubMed 2439802]
92. Larrey D, Henrion J, Heller F, et al. Metoprolol-induced hepatitis: rechallenge and drug oxidation phenotyping. *Ann Intern Med*. 1988;108(1):67-68. doi:10.7326/0003-4819-108-1-67 [PubMed 3337519]
93. Lederballe Pedersen O, Mikkelsen E, Lanng Nielsen J, Christensen NJ. Abrupt withdrawal of beta-blocking agents in patients with arterial hypertension. Effect on blood pressure, heart rate and plasma catecholamines and prolactin. *Eur J Clin Pharmacol*. 1979;15(3):215-217. doi:10.1007/BF00563108 [PubMed 37093]
94. Lee R. Atrial fibrillation and flutter after cardiac surgery. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 11, 2020.
95. Levine GN, Steinke EE, Bakaeen FG, et al; American Heart Association Council on Clinical Cardiology; Council on Cardiovascular Nursing; Council on Cardiovascular Surgery and Anesthesia; Council on Quality of Care and Outcomes Research. Sexual activity and cardiovascular disease: a scientific statement from the American Heart Association. *Circulation*. 2012;125(8):1058-1072. doi:10.1161/CIR.0b013e3182447787 [PubMed 22267844]
96. Liedholm H, Melander A, Bitzén PO, et al. Accumulation of Atenolol and Metoprolol in Human Breast Milk. *Eur J Clin Pharmacol*. 1981;20(3):229-231. [PubMed 7286041]
97. Lindeberg S, Lundborg P, Regårdh CG, et al. Disposition of the Adrenergic Blocker Metoprolol and its Metabolite OH-Metoprolol in Maternal Plasma, Amniotic Fluid and Capillary Blood of the Neonate. *Eur J Clin Pharmacol*. 1987;33(4):363-368. [PubMed 3443141]

98. Lipworth B, Wedzicha J, Devereux G, Vestbo J, Dransfield MT. Beta-blockers in COPD: time for reappraisal. *Eur Respir J*. 2016;48(3):880-888. doi:10.1183/13993003.01847-2015 [PubMed 27390282]
99. Liu X, Lou X, Cheng X, Meng Y. Impact of metoprolol treatment on mental status of chronic heart failure patients with neuropsychiatric disorders. *Drug Des Devel Ther*. 2017;11:305-312. doi:10.2147/DDDT.S124497 [PubMed 28182127]
100. López-Sendón J, Swedberg K, McMurray J, et al. Expert consensus document on beta-adrenergic receptor blockers. *Eur Heart J*. 2004;25(15):1341-1362. doi:10.1016/j.ehj.2004.06.002 [PubMed 15288162]
101. Lopressor (metoprolol tartrate) oral solution [prescribing information]. East Windsor, NJ: Validus Pharmaceuticals LLC; April 2025.
102. Lopressor (metoprolol tartrate, USP) tablets [prescribing information]. Parsippany, NJ: Validus Pharmaceuticals LLC; September 2023.
103. Lopressor (metoprolol) [product monograph]. Dorval, Quebec, Canada: Novartis Pharmaceuticals Canada Inc; April 2020.
104. Lopressor SR (metoprolol) [product monograph]. Dorval, Quebec, Canada: Novartis Pharmaceuticals Canada Inc; February 2022.
105. Lu HT, Kam J, Nordin RB, et al. Beta-blocker use and risk of symptomatic bradyarrhythmias: a hospital-based case-control study. *J Geriatr Cardiol*. 2016;13(9):749-759. doi:10.11909/j.issn.1671-5411.2016.09.009 [PubMed 27899939]
106. Magee LA, Smith GN, Bloch C, et al. Guideline No. 426: hypertensive disorders of pregnancy: diagnosis, prediction, prevention, and management. *J Obstet Gynaecol Can*. 2022;44(5):547-571.e1. doi:10.1016/j.jogc.2022.03.002 [PubMed 35577426]
107. Mayer S, Hillis LD. Prinzmetal's variant angina. *Clin Cardiol*. 1998;21(4):243-246. [PubMed 9562933]
108. McAinsh J, Cruickshank JM. Beta-blockers and central nervous system side effects. *Pharmacol Ther*. 1990;46(2):163-197. doi:10.1016/0163-7258(90)90092-g [PubMed 1969642]
109. Melander A, Danielson K, Scherstén B, Wåhlin E. Enhancement of the bioavailability of propranolol and metoprolol by food. *Clin Pharmacol Ther*. 1977;22(1):108-112. [PubMed 872491]
110. Metoprolol succinate extended-release capsules [prescribing information]. New Brunswick, NJ: Sun Pharmaceutical Industries; January 2018.
111. Metoprolol succinate extended-release tablets [prescribing information]. Södertälje, Sweden: AstraZeneca; March 2006.
112. Metoprolol tartrate injection USP [product monograph]. Mississauga, Ontario, Canada: Juno Pharmaceuticals Corp; July 2023.
113. Metoprolol tartrate injection ampuls [prescribing information]. Lehi, UT: Civica Inc; December 2020.
114. Metoprolol tartrate injection vials [prescribing information]. Deerfield, IL: Baxter Healthcare Corp; May 2024.
115. Metoprolol tartrate tablets [prescribing information]. Morgantown, WV: Mylan Pharmaceuticals; March 2015.
116. Metoprolol tartrate tablets, USP [product monograph]. Brampton, Ontario, Canada: Sanis Health Inc; April 2022.
117. Mills GA, Horn JR. Beta-blockers and glucose control. *Drug Intell Clin Pharm*. 1985;19(4):246-251. doi:10.1177/106002808501900401 [PubMed 2861072]

118. Morales DR, Jackson C, Lipworth BJ, Donnan PT, Guthrie B. Adverse respiratory effect of acute β -blocker exposure in asthma: a systematic review and meta-analysis of randomized controlled trials. *Chest*. 2014;145(4):779-786. doi:10.1378/chest.13-1235 [PubMed 24202435]
119. Morales DR, Lipworth BJ, Donnan PT, Jackson C, Guthrie B. Respiratory effect of beta-blockers in people with asthma and cardiovascular disease: population-based nested case control study. *BMC Med*. 2017;15(1):18. doi:10.1186/s12916-017-0781-0 [PubMed 28126029]
120. Morselli PL, Boutroy MJ, Bianchetti G, et al. Pharmacokinetics of Antihypertensive Drugs in the Neonatal Period. *Dev Pharmacol Ther*. 1989;13(2-4):190-198. [PubMed 2693002]
121. Muller G, Deal BJ, Strasburger JF, et al. Usefulness of Metoprolol for Unexplained Syncope and Positive Response to Tilt Testing in Young Persons. *Am J Cardiol*. 1993;71(7):592-595. [PubMed 8438747]
122. Myers MG, Thiessen JJ. Metoprolol kinetics and dose response in hypertensive patients. *Clin Pharmacol Ther*. 1980;27(6):756-762. doi:10.1038/clpt.1980.106 [PubMed 7379442]
123. National High Blood Pressure Education Program Working Group on High Blood Pressure in Children and Adolescents. The fourth report on the diagnosis, evaluation, and treatment of high blood pressure in children and adolescents. *Pediatrics*. 2004;114(suppl 2):555-576. [PubMed 15286277]
124. National Institutes of Health, National Heart, Lung, and Blood Institute. Expert panel on integrated guidelines for cardiovascular health and risk reduction in children and adolescents. Clinical Practice Guidelines, 2011. http://www.nhlbi.nih.gov/guidelines/cvd_ped/peds_guidelines_full.pdf.
125. Nattel S, Rangno RE, Van Loon G. Mechanism of propranolol withdrawal phenomena. *Circulation*. 1979;59(6):1158-1164. doi:10.1161/01.cir.59.6.1158 [PubMed 436208]
126. Neumar RW, Otto CW, Link MS, et al. Part 8: adult advanced cardiovascular life support: 2010 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care [published corrections appear in *Circulation*. 2011;123(6):e236; *Circulation*. 2013;128(25):e480]. *Circulation*. 2010;122(18)(suppl 3):S729-S767. doi: 10.1161/CIRCULATIONAHA.110.970988. [PubMed 20956224]
127. Nguyen DL, Wittich CM. Metoprolol-induced lichenoid dermatitis. *J Gen Intern Med*. 2011;26(11):1379-1380. doi:10.1007/s11606-011-1742-5 [PubMed 21614631]
128. Nicolai MP, Liem SS, Both S, et al. A review of the positive and negative effects of cardiovascular drugs on sexual function: a proposed table for use in clinical practice. *Neth Heart J*. 2014;22(1):11-19. doi:10.1007/s12471-013-0482-z [PubMed 24155101]
129. Nicpoń KW. Headache induced by oral intake of metoprolol in a patient with chronic and limited cardiac tamponade [in Polish]. *Neurol Neurochir Pol*. 2006;40(4):342-346. [PubMed 16967357]
130. Nikolaidou T, Ghosh JM, Clark AL. Outcomes related to first-degree atrioventricular block and therapeutic implications in patients with heart failure. *JACC Clin Electrophysiol*. 2016;2(2):181-192. doi:10.1016/j.jacep.2016.02.012 [PubMed 29766868]
131. Olsson G, Wikstrand J, Warnold I, et al. Metoprolol-induced reduction in postinfarction mortality: pooled results from five double-blind randomized trials. *Eur Heart J*. 1992;13(1):28-32. [PubMed 1533587]
132. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med*. 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]
133. O'Marcaigh AS, MacLellan-Tobert SG, Porter CJ. Tilt-Table Testing and Oral Metoprolol Therapy in Young Patients With Unexplained Syncope. *Pediatrics*. 1994;93(2):278-283. [PubMed 8121741]
134. Ommen SR, Ho CY, Asif IM, et al. 2024 AHA/ACC/AMSSM/HRS/PACES/SCMR guideline for the management of hypertrophic cardiomyopathy: a report of the American Heart Association.

- tion/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation*. 2024;149(23):e1239-e1311. doi:10.1161/CIR.0000000000001250 [PubMed 38718139]
135. Osmonov D, Erdinler I, Ozcan KS, et al. Management of patients with drug-induced atrioventricular block. *Pacing Clin Electrophysiol*. 2012;35(7):804-810. doi:10.1111/j.1540-8159.2012.03410.x [PubMed 22530749]
 136. Ozaydin M, Icli A, Yucel H, et al. Metoprolol vs. carvedilol or carvedilol plus N-acetyl cysteine on post-operative atrial fibrillation: a randomized, double-blind, placebo-controlled study. *Eur Heart J*. 2013;34(8):597-604. doi:10.1093/eurheartj/ehs423 [PubMed 23232844]
 137. Page RL, Joglar JA, Caldwell MA, et al. 2015 ACC/AHA/HRS guideline for the management of adult patients with supraventricular tachycardia: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society [published correction appears in *J Am Coll Cardiol*. 2016;68(25):2922-2923]. *J Am Coll Cardiol*. 2016;67(13):e27-e115. [PubMed 26409259]
 138. Panchal AR, Bartos JA, Cabañas JG, et al; Adult Basic and Advanced Life Support Writing Group. Part 3: adult basic and advanced life support: 2020 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2020;142(16)(suppl 2):S366-S468. doi:10.1161/CIR.0000000000000916 [PubMed 33081529]
 139. Park MK, Salamat M. Appendix E: Drugs used in pediatric cardiology. In: *Park's Pediatric Cardiology for Practitioners*. 7th ed. Elsevier Health Sciences; 2021: 477-492.
 140. Patakas D, Argiropoulou V, Louridas G, Tsara V. Beta-blockers in bronchial asthma: effect of propranolol and pindolol on large and small airways. *Thorax*. 1983;38(2):108-112. doi:10.1136/thx.38.2.108 [PubMed 6344309]
 141. Patel S, Kim S, Allen C. Metoprolol-induced pemphigus-like reaction. *Clin Adv Periodontics*. 2019;9(1):24-28. doi:10.1002/cap.10044 [PubMed 31490034]
 142. Perry A, Sparling JD, Pennington M. Bullous pemphigoid following therapy with an oral beta-blocker. *J Drugs Dermatol*. 2005;4(6):746-748. [PubMed 16302561]
 143. Philips C, Paramaguru R, Mahadevan P, Ravindranath J, Augustine P. Metoprolol-induced severe liver injury and successful management with therapeutic plasma exchange. *Cureus*. 2017;9(5):e1209. doi:10.7759/cureus.1209 [PubMed 28589058]
 144. Popp DA, Tse TF, Shah SD, Clutter WE, Cryer PE. Oral propranolol and metoprolol both impair glucose recovery from insulin-induced hypoglycemia in insulin-dependent diabetes mellitus. *Diabetes Care*. 1984;7(3):243-247. doi:10.2337/diacare.7.3.243 [PubMed 6376017]
 145. Pringsheim T, Davenport W, Mackie G, et al. Canadian Headache Society guideline for migraine prophylaxis. *Can J Neurol Sci*. 2012;39(2 Suppl 2):S1-S9. [PubMed 22683887]
 146. Proclemer A, Gradnik R, Savonitto S, Feruglio GA. Electrophysiological effects of bisoprolol. *Eur Heart J*. 1987;8(suppl M):81-85. doi:10.1093/eurheartj/8.suppl_m.81 [PubMed 2897302]
 147. Prystowsky EN. The effects of slow channel blockers and beta blockers on atrioventricular nodal conduction. *J Clin Pharmacol*. 1988;28(1):6-21. doi:10.1002/j.1552-4604.1988.tb03095.x [PubMed 2450898]
 148. Psaty BM, Koepsell TD, Wagner EH, LoGerfo JP, Inui TS. The relative risk of incident coronary heart disease associated with recently stopping the use of beta-blockers. *JAMA*. 1990;263(12):1653-1657. [PubMed 1968518]
 149. Ranchord AM, Spertus JA, Buchanan DM, Gosch KL, Chan PS. Initiation of β -blocker therapy and depression after acute myocardial infarction. *Am Heart J*. 2016;174:37-42. doi:10.1016/j.ahj.2015.11.018 [PubMed 26995368]
 150. Rangno RE, Langlois S. Comparison of withdrawal phenomena after propranolol, metoprolol and pindolol. *Br J Clin Pharmacol*. 1982;13(Suppl 2):345S-351S. doi:10.1111/j.1365-2125.1982.tb01939.x [PubMed 6125187]

151. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309 [PubMed 40014670]
152. Refer to manufacturer's labeling.
153. Regardh CG, Borg KO, Johansson R, Johnsson G, Palmer L. Pharmacokinetic studies on the selective beta₁-receptor antagonist metoprolol in man. *J Pharmacokinet Biopharm*. 1974;2(4):347-364. [PubMed 4155762]
154. Regardh CG, Johnsson G. Clinical pharmacokinetics of metoprolol. *Clin Pharmacokinet*. 1980;5(6):557-569. [PubMed 7002420]
155. Regårdh CG, Jordö L, Ervik M, Lundborg P, Olsson R, Rönn O. Pharmacokinetics of metoprolol in patients with hepatic cirrhosis. *Clin Pharmacokinet*. 1981;6(5):375-388. doi:10.2165/00003088-198106050-00004 [PubMed 7333059]
156. Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, et al. 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J*. 2018;39(34):3165-3241. [PubMed 30165544]
157. Riva-Metoprolol-L (metoprolol tartrate, USP) [product monograph]. Blainville, Quebec, Canada: Laboratoire Riva Inc; November 2020.
158. Rosendorff C, Lackland DT, Allison M, et al; American Heart Association, American College of Cardiology, and American Society of Hypertension. Treatment of hypertension in patients with coronary artery disease: A scientific statement from the American Heart Association, American College of Cardiology, and American Society of Hypertension. *J Am Soc Hypertens*. 2015;9(6):453-498. doi:10.1016/j.jash.2015.03.002 [PubMed 25840695]
159. Rosenson RS, Reeder GS, Kennedy HL. Acute myocardial infarction: role of beta blocker therapy. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 4, 2019.
160. Ross DS, Burch HB, Cooper DS, et al. 2016 American Thyroid Association guidelines for diagnosis and management of hyperthyroidism and other causes of thyrotoxicosis [published correction appears in *Thyroid*. 2017;27(11):1462]. *Thyroid*. 2016;26(10):1343-1421. doi:10.1089/thy.2016.0229 [PubMed 27521067]
161. Rössner S, Weiner L. Atenolol and metoprolol: comparison of effects on blood pressure and serum lipoproteins, and side effects. *Eur J Clin Pharmacol*. 1983;24(5):573-577. doi:10.1007/BF00542203 [PubMed 6873132]
162. Ryu RJ, Eyal S, Easterling TR, et al. Pharmacokinetics of metoprolol during pregnancy and lactation. *J Clin Pharmacol*. 2016;56(5):581-589. [PubMed 26461463]
163. Salpeter S, Ormiston T, Salpeter E. Cardioselective beta-blockers for reversible airway disease. *Cochrane Database Syst Rev*. 2002;(4):CD002992. doi:10.1002/14651858.CD002992 [PubMed 12519582]
164. Sandström B, Lindeberg S, Lundborg P, Regardh CG. Disposition of the adrenergic blocker metoprolol in the late pregnant women, the amniotic fluid, the cord blood and the neonate. *Clin Exp Hypertens B*. 1983;2(1):75-82. [PubMed 6135523]
165. Scheer FA, Morris CJ, Garcia JL, et al. Repeated melatonin supplementation improves sleep in hypertensive patients treated with beta-blockers: a randomized controlled trial. *Sleep*. 2012;35(10):1395-1402. doi:10.5665/sleep.2122 [PubMed 23024438]
166. Schellenberg R, Lichtenthal A, Wöhling H, Graf C, Brixius K. Nebivolol and metoprolol for treating migraine: an advance on beta-blocker treatment? *Headache*. 2008;48(1):118-125. doi:10.1111/j.1526-4610.2007.00785.x. [PubMed 18184294]

167. Schier JG, Howland MA, Hoffman RS, Nelson LS. Fatality from administration of labetalol and crushed extended release nifedipine. *Ann Pharmacother.* 2003;37(10):1420-1423. doi:10.1345/aph.1D091 [PubMed 14519033]
168. Schwedt TJ, Garza I. Preventive treatment of episodic migraine in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 21, 2022
169. Seidl K, Hauer B, Schwick NG, Zahn R, Senges J. Comparison of metoprolol and sotalol in preventing ventricular tachyarrhythmias after the implantation of a cardioverter/defibrillator. *Am J Cardiol.* 1998;82(6):744-748. [PubMed 9761084]
170. Seiler KU, Schuster KJ, Meyer GJ, Niedermayer W, Wassermann O. The pharmacokinetics of metoprolol and its metabolites in dialysis patients. *Clin Pharmacokinet.* 1980;5(2):192-198. doi:10.2165/00003088-198005020-00006 [PubMed 7363534]
171. Semet M, Paci M, Saïas-Magnan J, et al. The impact of drugs on male fertility: a review. *Andrology.* 2017;5(4):640-663. doi:10.1111/andr.12366 [PubMed 28622464]
172. Shaddy RE, Tani LY, Gidding SS, et al. Beta-Blocker Treatment of Dilated Cardiomyopathy With Congestive Heart Failure in Children: A Multi-Institutional Experience. *J Heart Lung Transplant.* 1999;18(3):269-274. [PubMed 10328154]
173. Shand DG, Wood AJ. Propranolol withdrawal syndrome - why?. *Circulation.* 1978;58(2):202-203. doi:10.1161/01.cir.58.2.202 [PubMed 668066]
174. Shehab N, Lewis CL, Streetman DD, Donn SM. Exposure to the pharmaceutical excipients benzyl alcohol and propylene glycol among critically ill neonates. *Pediatr Crit Care Med.* 2009;10(2):256-259. [PubMed 19188870]
175. Silberstein SD, Holland S, Freitag F, et al; Quality Standards Subcommittee of the American Academy of Neurology and the American Headache Society. Evidence-based guideline update: pharmacologic treatment for episodic migraine prevention in adults: report of the Quality Standards Subcommittee of the American Academy of Neurology and the American Headache Society [published correction appears in *Neurology.* 2013;80(9):871]. *Neurology.* 2012;78(17):1337-1345. [PubMed 22529202]
176. Silvestri A, Galetta P, Cerquetani E, et al. Report of erectile dysfunction after therapy with beta-blockers is related to patient knowledge of side effects and is reversed by placebo. *Eur Heart J.* 2003;24(21):1928-1932. doi:10.1016/j.ehj.2003.08.016 [PubMed 14585251]
177. Simons M, Breall JA. Overview of the non-acute management of unstable angina and non-ST elevation myocardial infarction. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 13, 2020.
178. Sirois FJ. Visual hallucinations and metoprolol. *Psychosomatics.* 2006;47(6):537-538. doi:10.1176/appi.psy.47.6.537 [PubMed 17116960]
179. Smith SC Jr, Benjamin EJ, Bonow RO, et al. AHA/ACC Secondary Prevention and Risk Reduction Therapy for Patients With Coronary and Other Atherosclerotic Vascular Disease: 2011 Update: A Guideline From the American Heart Association and American College of Cardiology Foundation. *Circulation.* 2011;124(22):2458-2473. [PubMed 22052934]
180. Snyder S. Metoprolol-induced polymyalgia-like syndrome. *Ann Intern Med.* 1991;114(1):96-97. doi:10.7326/0003-4819-114-1-96_2 [PubMed 1983947]
181. Steiner TJ, Jensen R, Katsarava Z, et al. Aids to management of headache disorders in primary care (2nd edition): on behalf of the European Headache Federation and Lifting the Burden: the Global Campaign against Headache. *J Headache Pain.* 2019;20(1):57. doi:10.1186/s10194-018-0899-2 [PubMed 31113373]
182. Tangeman HJ, Patterson JH. Extended-release metoprolol succinate in chronic heart failure. *Ann Pharmacother.* 2003;37(5):701-710. [PubMed 12708950]

183. Tatu AL, Nwabudike LC. Metoprolol-associated onset of psoriatic arthropathy. *Am J Ther.* 2017;24(3):e370-e371. doi:10.1097/MJT.0000000000000560 [PubMed 28221167]
184. Terentes-Printzios D, Ioakeimidis N, Rokkas K, Vlachopoulos C. Interactions between erectile dysfunction, cardiovascular disease and cardiovascular drugs. *Nat Rev Cardiol.* 2022;19(1):59-74. doi:10.1038/s41569-021-00593-6 [PubMed 34331033]
185. Tisdale JE, Chung MK, Campbell KB, et al. Drug-induced arrhythmias: a scientific statement from the American Heart Association. *Circulation.* 2020;142(15):e214-e233. doi:10.1161/CIR.0000000000000900 [PubMed 32929996]
186. Toprol XL (metoprolol succinate) extended-release tablet [prescribing information]. Parsippany, NJ: Melinta Therapeutics; February 2025.
187. Tsujimoto T, Sugiyama T, Shapiro MF, Noda M, Kajio H. Risk of cardiovascular events in patients with diabetes mellitus on β -blockers. *Hypertension.* 2017;70(1):103-110. doi:10.1161/HYPERTENSIONAHA.117.09259 [PubMed 28559400]
188. van den Berg G, van Steveninck F, Gubbens-Stibbe JM, Schoemaker HC, de Boer AG, Cohen AF. Influence of food on the bioavailability of metoprolol from an OROS system; a study in healthy volunteers. *Eur J Clin Pharmacol.* 1990;39(3):315-316. [PubMed 2257873]
189. Viigimaa M, Vlachopoulos C, Doumas M, et al; European Society of Hypertension Working Group on Sexual Dysfunction. Update of the position paper on arterial hypertension and erectile dysfunction. *J Hypertens.* 2020;38(7):1220-1234. doi:10.1097/HJH.0000000000002382 [PubMed 32073535]
190. Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation.* 2023;148(9):e9-e119. doi:10.1161/CIR.0000000000001168 [PubMed 37471501]
191. Vouri SM, Morris EJ, Jiang X, et al. Evaluation of a beta-blocker-edema-loop diuretic prescribing cascade: a prescription sequence symmetry analysis. *Am J Hypertens.* 2022;35(7):601-609. doi:10.1093/ajh/hpac013 [PubMed 35106529]
192. Vue MH, Setter SM. Drug-induced glucose alterations part 1: drug-induced hypoglycemia. *Diabetes Spectrum,* 2011;24(3):171-177.
193. Wang S, Zou R, Cai H, et al. Heart rate and heart rate difference predicted the efficacy of metoprolol on postural tachycardia syndrome in children and adolescents. *J Pediatr.* 2020;224:110-114. doi:10.1016/j.jpeds.2020.05.017 [PubMed 32464225]
194. Wang YY, Han ZH, Wang YL, et al. Baseline left ventricular ejection fraction associated with symptom improvements in both children and adolescents with postural tachycardia syndrome under metoprolol therapy. *Chin Med J (Engl).* 2021;134(16):1977-1982. doi:10.1097/CM9.0000000000001698 [PubMed 34387611]
195. Weber MA, Schiffrin EL, White WB, et al, Clinical practice guidelines for the management of hypertension in the community: a statement by the American Society of Hypertension and the International Society of Hypertension. *J Clin Hypertens (Greenwich).* 2014;16(1):14-26. [PubMed 24341872]
196. Weersink RA, Bouma M, Burger DM, et al. Evidence-based recommendations to improve the safe use of drugs in patients with liver cirrhosis. *Drug Saf.* 2018;41(6):603-613. doi:10.1007/s40264-017-0635-x [PubMed 29330714]
197. Westerlund A. Central nervous system side-effects with hydrophilic and lipophilic beta-blockers. *Eur J Clin Pharmacol.* 1985;(28)(suppl):73-76. doi:10.1007/BF00543714 [PubMed 2865153]
198. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart

Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71(6):1269-1324. doi:10.1161/HYP.0000000000000066 [PubMed 29133354]

199. White R, Bradnam V; British Pharmaceutical Nutrition Group. *Handbook of Drug Administration via Enteral Feeding Tubes*. 3rd ed. Pharmaceutical Press; 2015.
200. Wikstrand J, Andersson B, Kendall MJ, Stanbrook H, Klibaner M. Pharmacokinetic considerations of formulation: extended-release metoprolol succinate in the treatment of heart failure. *J Cardiovasc Pharmacol*. 2003;41(2):151-157. [PubMed 12548073]
201. Wynn RL. Dental Nonsteroidal Anti-inflammatory Drugs and Prostaglandin-Based Drug Interactions, Part Two. *Gen Dent*. 1992;40(2):104, 106, 108. [PubMed 1354194]
202. Wynn RL. Epinephrine Interactions With Beta-Blockers. *Gen Dent*. 1994;42(1):16, 18. [PubMed 7911769]
203. Xu BW, Zhang QY, Li XY, et al. A predictive model of response to metoprolol in children and adolescents with postural tachycardia syndrome. *World J Pediatr*. 2023;19(4):390-400. doi:10.1007/s12519-022-00677-4 [PubMed 36781629]
204. Yilmaz MB, Turhan H, Akin Y, Kisacik HL, Korkmaz S. Beta-blocker-induced psoriasis: a rare side effect--a case report. *Angiology*. 2002;53(6):737-739. doi:10.1177/000331970205300617 [PubMed 12463630]
205. Yska JP, Wanders JTM, Odigie B, et al. Effect of roux-en-Y gastric bypass on the bioavailability of metoprolol from immediate and controlled release tablets: a single oral dose study before and after surgery. *Eur J Hosp Pharm*. 2020;27(e1):e19-e24. doi:10.1136/ejhpharm-2018-001804 [PubMed 32296500]
206. Zamir A, Hussain I, Ur Rehman A, et al. Clinical pharmacokinetics of metoprolol: a systematic review. *Clin Pharmacokinet*. 2022;61(8):1095-1114. doi:10.1007/s40262-022-01145-y [PubMed 35764772]
207. Zeltser D, Justo D, Halkin A, et al. Drug-induced atrioventricular block: prognosis after discontinuation of the culprit drug. *J Am Coll Cardiol*. 2004;44(1):105-108. doi:10.1016/j.jacc.2004.03.057 [PubMed 15234417]
208. Zimetbaum PJ, Wylie JV. Nonsustained ventricular tachycardia: Clinical manifestations, evaluation, and management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 4, 2019.

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